Distribution-Independent Tour-Length Estimation for the Traveling Salesman Problem Across Dimensions

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Abstract

We present the Geometrically Assisted Regression Tree (GART) 2.0, an improved estimator, to demonstrate that a learned model can give a reliable tour-length estimate with bounded complexity across dimensions and cases. Existing estimators either assume a known sampling density or depend on planar geometry that becomes unstable as dimension grows. GART 2.0 is a LightGBM regressor that predicts the instance-specific ratio $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ of tour length to minimum-spanning-tree (MST) length and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha}\, L_{\mathrm{MST}}$; no distribution label, no density, and no convex hull enters the computation. Trained on synthetic instances alone, it transfers to TSPLIB95 and, through a multidimensional-scaling embedding, to non-Euclidean instances given only as a distance matrix. On all four benchmark aggregates GART 2.0 leads every baseline scored there, from a seven-baseline roster, and on the multidimensional benchmark its dispersion falls monotonically in the two arguments the feature set was built to carry—growing instance size and growing dimension—out to a hundred dimensions, a value withheld from training entirely. We then score a known Held–Karp 1-tree lower bound, which consumes no training corpus and proves its own output, and use it to fix the boundary of the method. On the corpus aggregate the bound is the better instrument; that aggregate is mostly small instances, where the relaxation is nearly exact and nearly free, and read at fixed instance size the ordering turns over. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the plane it is strictly better than the bound on both axes from an ascent budget of 25 upward. The reason is structural: the bound pays a quadratic price in instance size for a fixed accuracy in every dimension, while the estimator's cost carries no accuracy term—so the estimator sharpens exactly where an exact solve becomes infeasible. Comparing GART 2.0's performance to classical estimators and bounds shows it has a niche in high-throughput tour-length estimation for large instances. However, it requires further refinement to perform better on non-Euclidean real-world data.

***K*eywords** Traveling Salesman Problem · Tour-length estimation · Distribution-independent · Arbitrary dimension

1 Introduction

The Traveling Salesman Problem (TSP) is NP-hard [Karp, 1972, Papadimitriou, 1977] and appears throughout logistics operations planning, network design, and resource allocation [Chien, 1992, Applegate et al., 2006]. It often enters as a subproblem of a larger optimization, where only its cost is needed. A facility-location decision, for example, requires the route cost of every candidate, so a full solve is paid once per candidate.

1.1 Background and Motivation

Last-mile delivery accounts for approximately 53% of total shipping costs by one industry estimate [Finmile], and the UPS ORION system is reported to save roughly 100 million miles and $300–400 million annually through route optimization [Holland et al., 2017]. In such systems orders arrive continuously and routes are long, so the cost of a full solve is paid many times over. An estimator replaces that solve wherever only the relative cost of candidate routes matters.

The same estimation problem recurs wherever a TSP appears as a subproblem, and it rarely arrives in the plane. Radiation hybrid mapping orders genetic markers by a TSP over an abstract similarity metric [Agarwala et al., 2000]. Mobile-sink routing in wireless sensor networks contains a TSP through the data-collection locations [Yuan and Peng, 2010]. UAV structural inspection places viewpoints around a three-dimensional structure and solves a TSP through them [Bircher et al., 2016]. Large-scale routing methods partition an instance into sub-TSPs and compare candidate partitions [Mariescu-Istodor and Fränti, 2021], and Pan et al. [2023] learn a hierarchical policy that repeatedly selects node subsets and routes through them. Each of these planners needs the cheaper option predicted as cheaper, so the estimator must be accurate and rank-preserving.

Most classical tour-length estimators, however, are built for low-dimensional Euclidean inputs and depend on geometric quantities—most often convex-hull area—that become unstable or intractable as dimension grows [Chazelle, 1993]. An estimator for these settings must be defined for arbitrary d without building a hull. A distance matrix admits a Euclidean representation of bounded distortion once enough dimensions are allowed [Bourgain, 1985], so an estimator that stays accurate at those dimensions can consume the embedding.

1.2 Limitations of the Current Estimators

No published estimator reaches high-dimensional or non-Euclidean instances without losing calibration or requiring an embedding. The Held–Karp 1-tree bound reaches both directly, but it is a certificate, not a calibrated estimator. The Beardwood–Halton–Hammersley (BHH) theorem [Beardwood et al., 1959] is stated for arbitrary dimension, but it is asymptotic in n and its calibrating constants and density integral are not recoverable from an arbitrary finite point set. Published constants exist for $d = 2$ [Johnson et al., 1996] and $d = 3$ [Percus and Martin, 1996], and above that only a large-d approximation [Bertsimas and van Ryzin, 1990]. The estimators of Chien [1992] are planar and depend on particular depot and covering-region definitions. A parallel line of work is tuned to the 2D Euclidean case: Daganzo [1984b] gives a strip-based construction, Kwon et al. [1995] fit regression and neural models on rectangular-region predictors, and Çavdar and Sokol [2015] propose a distribution-free estimator built from per-axis spread statistics about the covering rectangle. Space-filling heuristics such as the planar construction of Bartholdi and Platzman [1982] instead build a tour outright in $O(n \log n)$. Embedding a non-Euclidean instance therefore returns the classical estimators above to the dimensions where their geometric inputs are unstable. We survey Chien [1992], Daganzo [1984b] and Kwon et al. [1995] without scoring them.

Recent neural approaches address the same gap from the learning side. Varol et al. [2024] report sub-1% deviation on standard distributions by combining a multilayer perceptron (MLP) with domain-knowledge features, but their pipeline requires solution-level features, which are computationally expensive and challenging to generalize across all instance sizes. Kou et al. [2022] use the standard deviation of random feasible tour lengths as a single-feature, distribution-independent predictor on Euclidean and non-Euclidean instances, but the repeated tour sampling raises its per-instance cost. Kou et al. [2024] extend this principle beyond the TSP to the vehicle-routing (VRP), MST, and matching problems, again from the standard deviation of random feasible solutions. We exclude that sampling cost on the same grounds as solution-level features. The closest methodological neighbor is Kou et al. [2023], which pairs a Sammon-map embedding with a distribution-independent estimator on non-Euclidean TSPLIB95 instances. Our pipeline uses classical metric MDS for the embedding and computes MST-topology features on the original distance matrix, not on the embedded coordinates.

A separate classical line bounds the tour instead of estimating it. Held and Karp [1970, 1971] relax the Hamiltonian-cycle constraint to a *1-tree*, a spanning tree on all but one node plus any two edges at that node. The minimum 1-tree under any vector of node penalties yields a proven lower bound on the optimal tour, so an ascent over penalties stopped anywhere returns a certificate. The bound uses no training data and no distributional assumption, and it does not invoke the triangle inequality: it is defined for any symmetric distance matrix, including the non-Euclidean instances Section 6 reaches only through an embedding. Its cost is quadratic in n per ascent iteration. In the plane and at $d = 3$, Section 5 shows that GART 2.0 beats this bound at large n: feature extraction is subquadratic while the ascent is quadratic in every dimension, and the accuracy it gives up is small.

1.3 Contribution

We present the Geometrically Assisted Regression Tree (GART) 2.0, a LightGBM regressor [Ke et al., 2017] that predicts a per-instance scaling factor $\hat{\alpha}$ rather than applying a fixed constant. It is distribution-independent: no distribution label and no parametric density is supplied at inference, although accuracy varies by distribution type. Çavdar and Sokol [2015] and Kou et al. [2022, 2023] share that property, but remain tied to 2D coordinate geometry or to per-instance sampling. The contribution of GART 2.0 is to be distribution-independent *and* dimension-scalable at once: all 31 features are defined for arbitrary d through MST topology, centroid dispersion and coordinate ranges, no feature requires a convex hull, and the same feature set applies unchanged as both dimension and instance size grow.

GART 2.0 gives stronger accuracy and precision than classical tour-length estimators across a broad range of cases, at higher computational cost. However, compared with the known lower bound, it is the more efficient instrument on large instances (Section 5).

The Held–Karp 1-tree bound is defined on every instance in the corpus and on some that GART 2.0 declines. On the corpus aggregate the bound is the better instrument. Stratified by size as well as by dimension (Table 6), the ordering turns over above a few hundred nodes, and the mechanism is the complexity separation of Section 5.2.

2 Theoretical Foundations

The ratio $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ varies from instance to instance. Let $\mathcal{G} = (V, E)$ be a complete undirected graph on the node set $V = \{1, \ldots, n\}$ with non-negative edge costs c_{ij}, typically the Euclidean distance between points in $\mathbb{R}^d$. Throughout, n is the number of nodes and d is the dimension of the coordinate space. The TSP asks for a minimum-cost Hamiltonian cycle, a connected 2-regular spanning subgraph [Dantzig et al., 1954, Applegate et al., 2006]. Relaxing 2-regularity to connectivity alone gives the minimum spanning tree (MST).

Prim's algorithm constructs one in $O(n^2)$ on a complete graph [Prim, 1957], and in $\mathbb{R}^d$ the pairwise distances cost $O(n^2 d)$, linear in d. Removing the heaviest edge from any tour leaves a spanning tree, so $L_{\mathrm{MST}} \leq L_{\mathrm{TSP}}$. The double-tree heuristic [Rosenkrantz et al., 1977] builds a feasible tour of length at most $2\,L_{\mathrm{MST}}$ on any instance obeying the triangle inequality. Together these give

$$L_{\mathrm{MST}} \leq L_{\mathrm{TSP}} \leq 2 \cdot L_{\mathrm{MST}}. \tag{1}$$

The upper constant cannot be reduced in general: for collinear point sets the optimal tour must traverse the line to its far end and return, so $L_{\mathrm{TSP}} = 2\,L_{\mathrm{MST}}$ exactly.

A tighter known lower bound comes from the Held–Karp relaxation. Fix a node s. A *1-tree* is a spanning tree on $V \setminus \{s\}$ together with any two edges incident to s. Every Hamiltonian tour is a 1-tree, so the minimum 1-tree relaxes the tour problem. Introduce node potentials $\pi \in \mathbb{R}^n$ and transformed costs $c^{\pi}_{ij} = c_{ij} + \pi_i + \pi_j$. Under c^{π} every tour's cost shifts by exactly $2\sum_i \pi_i$, because every node of a tour has degree two, and for *any* real π

$$w(\pi) \;=\; L_{\mathrm{1tree}}(\pi) \;-\; 2\sum_{i \in V} \pi_i \;\leq\; L_{\mathrm{TSP}}, \tag{2}$$

which is the Held–Karp bound [Held and Karp, 1970, 1971]. The map w is concave and piecewise linear, a subgradient at π is $g_i = \deg_i(\pi) - 2$, and subgradient ascent over π approaches $\max_{\pi} w(\pi)$. Each iteration builds its spanning tree on the perturbed costs directly. The ascent is not monotone in w, so the scored quantity at budget k is the incumbent

$\hat{w}_k = \max_{j \le k} w(\pi_j)$. Eq. (2) needs neither $\pi \ge 0$ nor $c^\pi \ge 0$, so $\hat{w}_k$ is a *certificate* at any budget. The bound consumes no training corpus.

The Christofides–Serdyukov algorithm [Christofides, 1976, Serdyukov, 1978] returns a tour within a factor $\frac{3}{2}$ of the *optimum*, and Karlin et al. [2021] reach a randomized $\frac{3}{2} - \epsilon$ approximation by a different construction. Both bound an algorithm's output against the optimal tour, so neither tightens the intrinsic ratio $L_{\mathrm{TSP}}/L_{\mathrm{MST}}$, which can still equal 2. The matching step costs $O(n^3)$ [Edmonds, 1965, Gabow, 1990, Cook and Rohe, 1999] against $O(n^2)$ for the MST, so the Christofides refinement is impractical as an estimation subroutine.

For symmetric metric instances we therefore treat $\alpha \in [1, 2]$ as an intrinsic, instance-dependent quantity to be predicted. Over our training split the standard deviation of α falls monotonically from 0.184 at $d = 2$ to 0.0762 at $d = 50$, while the cost of computing either quantity does not. For uniformly distributed points the TSP constant itself varies with dimension [Percus and Martin, 1996], and no published MST constant covers every d. A constant calibrated in the plane is therefore biased at higher d. GART 2.0 predicts α from the observed instance features instead.

3 Methodology: The Geometrically Assisted Regression Tree

GART 2.0 extends the estimator of Feldman [2025]. It predicts α and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha} L_{\mathrm{MST}}$. We constrain the regression target to $\alpha \in [1, 2]$, the metric-TSP interval implied by Eq. (1).

3.1 Feature Selection

3.1.1 Geometric and Centroid Descriptors

The geometric descriptors, such as bounding hypervolume and centroid distance, generalize to arbitrary d without a convex hull. They encode the scale information of the BHH volume term [Beardwood et al., 1959] and adapt centroid statistics from Çavdar and Sokol [2015]. Before evaluating them we rotate the coordinates into their principal-axis frame [Jolliffe, 2002], which makes the axis-aligned bounding-box descriptors independent of the input orientation. The rotation costs $O(nd^2 + d^3)$ and is dominated by dense MST construction whenever n is large relative to d. Appendix C lists the geometric group.

3.1.2 MST Edge and Topological Features

MST edge weights and the degree sequence are invariant to rotation, translation and coordinate-system choice. The MST's total length is not among the edge and topology statistics, because it is the multiplicand L_{MST} and withholding it keeps the learned quantity a scale-free ratio. The one input GART 2.0 adds to the 30-feature block it inherits is `greedy_nn_over_mst`, the length of an exact greedy nearest-neighbor tour divided by L_{MST}. A greedy tour is feasible, so its length is at least L_{TSP} and the ratio is a constructed upper bound on α. That ratio also gates coverage on the non-Euclidean path of Section 6, where the metric is given but coordinates are not. The estimator returns a status row instead of a prediction for any instance whose ratio falls outside $[1.035, 2.209]$, the range the ratio spans over the full 106,272-row corpus, training, validation and test together. The training split alone spans the narrower $[1.0465, 2.1295]$.

3.2 Dataset Generation

We built the corpus from four one-dimensional generators: Uniform, Normal, Clustered, and Correlated (Appendix B). Each axis of each instance draws its generator independently, so configurations are anisotropic and topology varies by axis. Over all 2,444,256 axes, 68.7% are uniform, 12.5% clustered, 12.5% normal, and 6.2% correlated. The multidimensional results below should be read against that mix.

The full feature corpus comprises 106,272 solved instances with node counts $n \in [5, 1000]$ and dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ (17 training-visible dimension values), plus a held-out extrapolation bucket at $d = 100$. Instances are partitioned into training (69,768), validation (19,584), and test (16,920) via a stratified split on $(d, n,$ grid size); all $d = 100$ instances are locked into the test partition.

For each instance, the generation pipeline ran Concorde [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017] and retained the shorter tour. The canonical feature table stores the retained tour cost but not solver-specific costs or Concorde certificates. We therefore call these values reference tour costs and do not claim that every stored row is independently auditable as optimal (Appendix E).

3.3 Model Architecture and Training

The estimator for α is a gradient-boosted decision tree (GBDT) ensemble [Friedman, 2001], implemented in LightGBM [Ke et al., 2017]. The learner uses squared-error regression on the logit-transformed target $z = \log\big((\alpha - 1)/(2 - \alpha)\big)$, and predictions are returned through its inverse $\alpha = 1 + (1 + e^{-z})^{-1}$. The bound $\alpha \in [1, 2]$ is therefore structural. The inverse transform cannot leave the open interval $(1, 2)$, so the collinear extreme of Eq. (1) is never attained. A clip to $[1, 2]$ remains in the released wrapper as a guard against non-finite inputs, and it is never binding. On the 16,920 multidimensional test predictions $\hat{\alpha}$ fell in $[1.03, 1.92]$. The hyperparameters are inherited and frozen. An Optuna search [Akiba et al., 2019] against in-distribution validation moved multidimensional MAPE by 0.011 percentage points while costing 0.16 points of TSPLIB MAPE, and we rejected it (Appendix I). We fit the model on the training split only, with no train–validation refit, and scored the test split once after that fit.

Two inputs, n and d, are constrained to be non-increasing, encoding that α falls toward 1 as an instance grows or the ambient dimension rises. The constraints are enforced inside the tree builder, so a sum of such trees cannot increase when either variable is swept alone. A probe over 1,000 held-out instances returns 100% non-increasing sweeps on both axes, against 8.40% of the dimension sweeps and 19.0% of the size sweeps for the same fit with the constraints removed. The probe overwrites one column of a real held-out instance and holds the other thirty at that instance's measured values, so a swept row corresponds to no realizable instance. It measures the shape of the learned function in one argument and says nothing about accuracy at those n and d, and a constant multiple of L_{MST} passes it trivially.

3.4 Feature Importance

By mean absolute SHAP magnitude [Lundberg and Lee, 2017] against the released booster, the MST dominance ratio contributes 26.5% of the total and `greedy_nn_over_mst` 23.9%. Node count and dimension jointly contribute 14.1%. Appendix D gives the group-level shares and the sample they are measured on.

A *cell* is one exact dimension and one size band of the held-out test split. Within cells the greedy-to-MST ratio holds 35.0% of the SHAP variance at $d = 2$, flat across every band, and the MST dominance ratio holds 23.9% to 27.5%. Within-cell fit in α space rises with dimension, from R^2 0.870 at $d = 2$ to 0.974 at the withheld $d = 100$. The generator mix explains 14.1% of the within-cell target at $d = 2$ and close to zero by $d = 100$. The geometric group carries scale. The MST edge statistics track the sampling distribution, and that dependence fades with dimension.

4 Benchmarking

Each results table reports the standard deviation of percent error (SDPE), the mean absolute percentage error (MAPE), the median absolute percentage error (the |Median| column), and the mean signed percent error (MSPE). Rows that predict a multiple of L_{MST} also carry $R^2_\alpha = 1 - \sum_i (\hat{\alpha}_i - \alpha_i)^2 / \sum_i (\alpha_i - \bar{\alpha})^2$, with $\bar{\alpha}$ the mean of α over the reported bucket. SDPE is the primary metric:

$$\mathrm{SDPE} = 100 \cdot \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (e_i - \bar{e})^2}, \qquad \bar{e} = \frac{1}{N} \sum_{i=1}^{N} e_i.$$

Here $e_i = (\hat{L}_i - L_i)/L_i$ is the signed relative error and N is the number of scored instances. Bold marks GART 2.0, not the best value in a column. Several baselines beat it in individual cells. Each SDPE carries a 95% bootstrap confidence interval, suppressed where a group holds fewer than three instances. The intervals are computed per estimator and ignore the pairing, so overlapping intervals establish nothing on their own. We test each comparison on the instances where both estimators produce a finite prediction, using the paired difference in absolute percent error with a 1,000-resample paired bootstrap and a Wilcoxon signed-rank test.

4.1 Benchmark Roster

We score estimators against their published domains. In the BHH form, $\mathcal{V}$ is the measure of the sampling region [Beardwood et al., 1959]: $\beta_2 = 0.7124$ [Johnson et al., 1996], $\beta_3 = 0.6979$ [Percus and Martin, 1996], and the conjectured large-d asymptotic is $\beta_d \approx \sqrt{d/(2\pi e)}$ for $d \geq 4$ [Bertsimas and van Ryzin, 1990]. In the Çavdar–Sokol form, A is the area of the rectangle covering the nodes, stdev_j is the standard deviation of the raw coordinates on axis j, and $\bar{c}_j$ and cstdev_j are the mean and standard deviation of the absolute distances from the nodes to that rectangle's midpoint line on axis j. The Çavdar–Sokol estimate is divided by the finite-n adjustment factor its source fits, held fixed outside the node range that fit covers. Both constant sets and the adjustment factor come from Çavdar [2014]. The asymptotic ratio's constant, $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}} \approx 1.1253$ [Johnson et al., 1996, Cortina-Borja and Robinson, 2000], comes from the literature; $\hat{\rho}(d, n)$ is fitted on the training split.

GART 1.0 [Feldman, 2025] is our prior model and the one learned estimator in the baseline count. It treats the lexicographically first unique coordinate as a depot-like reference for two features, although no depot is defined in these datasets. It differs from GART 2.0 in data, features, and training, so the comparison isolates no single causal change. The planar guarantee of Bartholdi and Platzman [1982] does not transfer to the Hilbert row's multidimensional variant (Table 1).

Table 1: Benchmark roster. D and N are the dimension and instance-size range each source fits or derives; a dash means the method reads only a distance matrix, so no dimension applies.

Estimator	**Formula**	**D**	**N**	**Cost**	**Source**
Classical region-based estimators					
BHH	$\beta_d\, n^{(d-1)/d} \mathcal{V}^{1/d}$, independent, uniformly distributed on a region of measure $\mathcal{V}$	$[2, \infty)$	$\to \infty$	$O(nd)$	Beardwood et al. [1959], Johnson et al. [1996], Bertsimas and van Ryzin [1990]
Çavdar–Sokol	$2.791\sqrt{n\, \mathrm{cstdev}_x \mathrm{cstdev}_y} + 0.2669\sqrt{n\, \mathrm{stdev}_x \mathrm{stdev}_y A/(\bar{c}_x \bar{c}_y)}$	$\{2\}$	$[100, 975]$	$O(nd)$	Çavdar [2014]
Constant-ratio references					
L_{MST} ($\alpha = 1$)	L_{MST}	–	$[3, \infty)$	$O(n^2 d)$	Lower bound of Eq. (1)
Asymptotic MST ratio	$(\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}})\, L_{\mathrm{MST}} \approx 1.1253\, L_{\mathrm{MST}}$, uniform limit	$\{2\}$	$\to \infty$	$O(n^2 d)$	Johnson et al. [1996], Cortina-Borja and Robinson [2000]
Calibrated MST ratio $\hat{\rho}(d, n)$	$\hat{\rho}(d, n)\, L_{\mathrm{MST}}$, one constant per trained cell	$[2, 50]$	$[5, 1000]$	$O(n^2 d)$	This work
Prior model and constructive heuristic					
GART 1.0	2D-only feature set, GBDT regression on α	$\{2\}$	$[3, \infty)$	$O(n^2)$	Feldman [2025]

Table 1 continued from the previous page

Estimator	Formula	D	N	Cost	Source
Custom Hilbert sort	Bare d-dimensional Hilbert ordering, per-axis normalized, order $h = 16$; returns a tour	$[2, \infty)$	$[3, \infty)$	$O(n \log n)$	After Bartholdi and Platzman [1982]
Known lower bound					
1-tree, Volgenant–Jonker step	incumbent $\hat{w}_k = \max_{j \leq k} w(\pi_j)$ over budget k; the schedule consults no upper bound	–	$[3, \infty)$	$\Theta(n^2 d + kn^2)$	Held and Karp [1970, 1971], Volgenant and Jonker [1982], Helsgaun [2000]
1-tree, Polyak step	the same incumbent; step $\gamma(\mathrm{UB} - w)/\|g\|^2$ against a constructive tour	–	$[3, \infty)$	$\Theta(n^2 d + kn^2)$	Held and Karp [1970], Polyak [1969]

4.2 Datasets

TSPLIB95's screened non-EUC_2D instances form the fourth benchmark (Section 6), and our generation pipeline produces the held-out synthetic sets (Appendix B). Non-metric instances stay out of every metric-dependent aggregate here.

4.2.1 Multidimensional Synthetic

The split shares the training generators, so it tests held-out sampling, not held-out geometry. Appendix B.3 gives the full (d, n) count grid (Table 9).

4.2.2 2D Diverse Synthetic

Each class in Table 2 probes a different assumption the classical 2D estimators make. Two of the thirteen sub-generators, Line Noise and `grid`, test geometry the training corpus does not contain (Appendix B.2).

Table 2: 2D benchmark dataset composition (2,580 instances).

Class	Generators	Count	Stress target
Isotropic [Çavdar and Sokol, 2015, Golden and Stewart, 1985]	Uniform, Normal, Triangular, Truncated Exp.	840	Control: $A_{\text{hull}} \approx A_{\text{box}}$.
Biased / Skewed [Çavdar and Sokol, 2015]	Squeezed, Uni-Tri, Tri-Squeezed, Correlated	840	Aspect-ratio distortion: $\sigma_x \neq \sigma_y$.
Geometric Struct. [Reinelt, 1991, Bossek et al., 2019]	Grid†, Boundary (hollow), X-Central	630	Lattice regularity, hollow voids.
Clustered [Golden and Stewart, 1985, Bossek et al., 2019]	k-Means ($k \in [5, 20]$)	60	Bimodal edge-weight distribution.
Line Noise†	Boundary-clipped $y = mx + b + \epsilon$; realized as a 2–4 segment union	210	Locally 1D structure; hull/box divergence.
Total		**2,580**	

† Not represented in training.
`Grid` is one of the three Geometric Struct. generators.

4.2.3 TSPLIB95 EUC_2D

The benchmark contains 78 EUC_2D instances with n from 51 (`eil51`) to 18,512 (`d18512`) [Reinelt, 1991]. Every 2D baseline scored here returns a value on all 78, the paper's common instance set.

4.3 Results

4.3.1 Multidimensional Benchmark

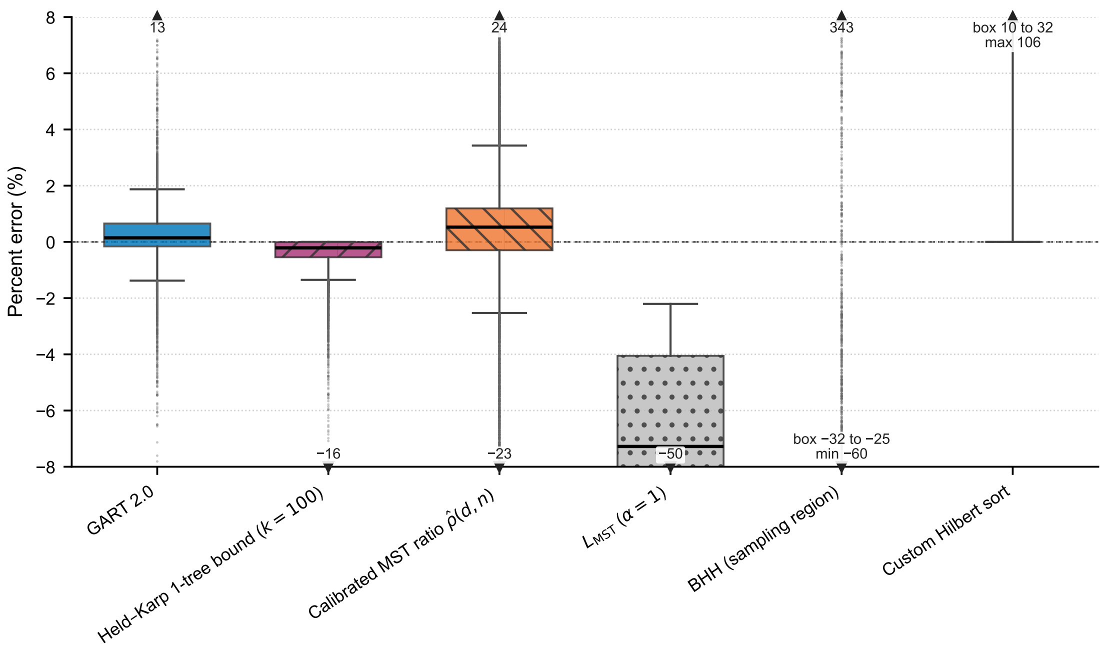

Figure 1: Percent error on the multidimensional benchmark ($N = 16{,}846$ per estimator).

GART 2.0 obtains 0.618% MAPE and 0.986% SDPE overall (Table 3), a factor of 2.9 on MAPE and 3.0 on SDPE over the strongest baseline that uses no learned model, the calibrated ratio $\hat{\rho}(d, n)$ at 1.82%/2.94%. That baseline is a lookup table of 102 constants, one per trained cell, fitted on the same training split. The multidimensional benchmark is 16,920 held-out instances, of which the 16,846 with a recoverable reference cost are scored (Appendix E). Figures 1, 3, and 4 draw the Held–Karp 1-tree bound at ascent budget $k = 100$. Several estimator boxes also fall on one side of the line, but only as a fitted tendency observed after the fact.

Table 3: Multidimensional benchmark results, by dimension band.

		GART 2.0		$\hat{\rho}(d,n)$		$\alpha = 1$		BHH		Hilbert	
Bucket	Count	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE
$d = 2$	647	**1.57**	**2.37**	4.05	6.19	19.1	9.78	18.9	30.9	29.7	15.8
$d \in [3, 5]$	1,944	**1.07**	**1.71**	2.53	4.05	15.0	7.67	26.8	24.1	35.7	15.7
$d \in [6, 10]$	3,238	**0.644**	**1.01**	1.79	2.98	11.9	6.46	27.6	10.3	32.5	13.8
$d \in [15, 25]$	1,942	**0.421**	**0.679**	1.40	2.46	9.40	5.98	27.0	6.17	24.6	11.7
$d \in [30, 50]$	3,228	**0.291**	**0.477**	1.29	2.31	8.02	5.94	28.2	4.34	17.1	7.99
$d = 100^{\dagger}$	5,847	**0.593**	**0.441**	1.78	2.21	6.73	5.93	29.9	2.76	10.4	4.60
Total	16,846	**0.618**	**0.986**	1.82	2.94	9.71	7.26	28.0	13.6	21.2	14.4

† Not represented in training.

SDPE falls monotonically with instance size and with dimension (Figure 2). The unseen $d = 100$ row extrapolates off the training grid: SDPE falls further there while MAPE rises.

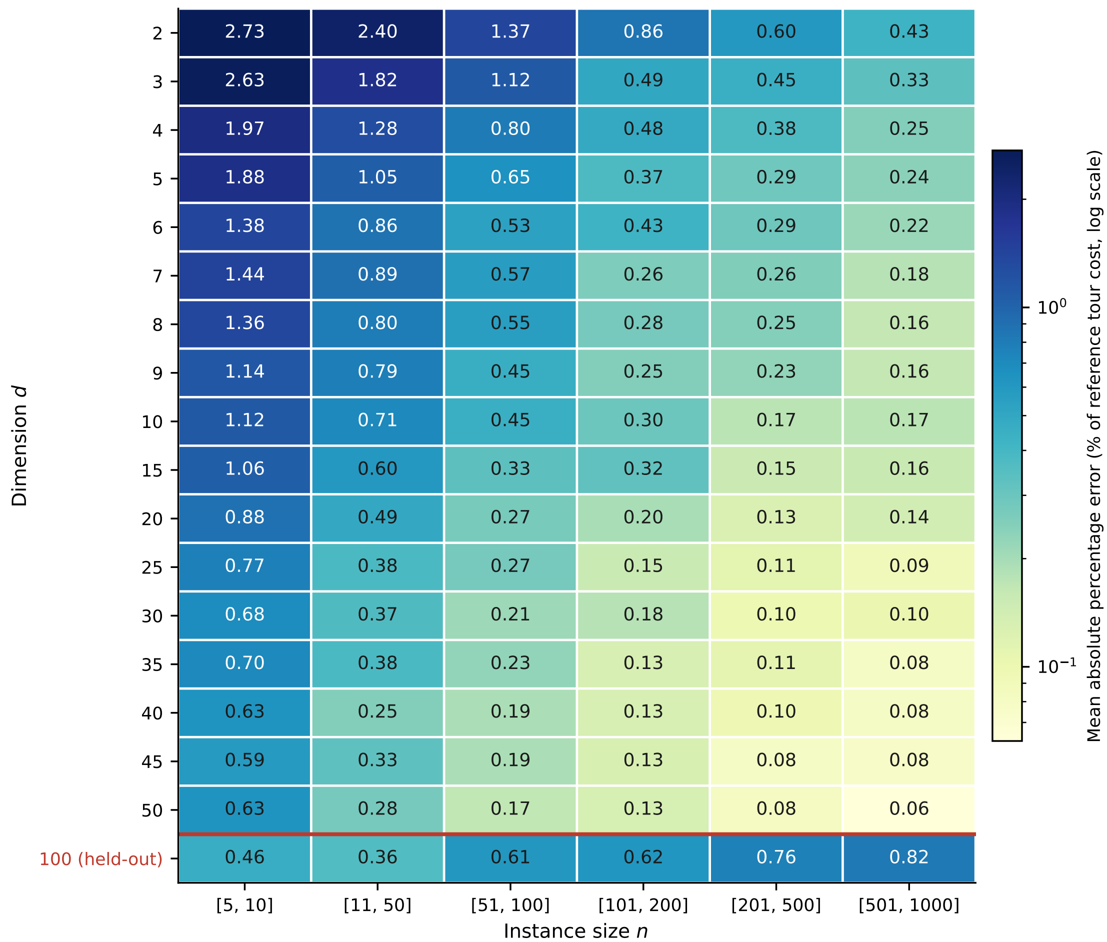

Figure 2: GART 2.0 MAPE by dimension and instance-size band; the red rule marks the held-out $d = 100$ row.

The constant dominates BHH's 28.0% error: for $d \geq 4$ the scored form substitutes a conjectured large-d asymptotic for β_d (Appendix F), and the row's signed error is negative at every $d \geq 3$, the direction an understated constant produces. The theorem's uniformity and large-n premises also fail here. Only 68.7% of axes are uniform, so the region measure G^d exceeds the effective support of the rest, visible as the positive signed error at $d = 2$. Instances at $n \leq 1000$ with $d \geq 4$ are far from the asymptotic regime the theorem describes, where the sample must fill the region. Supplying the exact region in place of the convex hull nonetheless cuts its MAPE from 39.1% to 28.0%, the share of the convex-hull form's error that is an input substitution. At $d > 3$ that hull is replaced by the coordinate bounding box (Appendix F), so the row would mix two region definitions across dimensions.

4.3.2 2D Diverse Benchmark

GART 2.0 obtains 2.89% MAPE and 4.68% SDPE overall (Table 4, Figure 3), roughly half the error of the strongest baseline, the calibrated ratio $\hat{\rho}(d, n)$ at 5.76%/9.38%. Its SDPE stays below 5.9% in every size bucket and falls to 2.75% in the $[501, 1000]$ bucket. The custom Hilbert sort has 30.9% MAPE against 14.2% SDPE, so its error is a near-constant positive offset, the signature a space-filling-curve tour should produce.

Every estimator degrades from the isotropic class to Line Noise. Near-collinear point sets drive α toward its upper bound of 2, and no estimator anchored on a typical ratio tracks that. Realized α averages 1.05 on `grid` against 1.56 on Line Noise, so the two unrepresented generators sit at opposite ends of the target range and the estimator errs toward

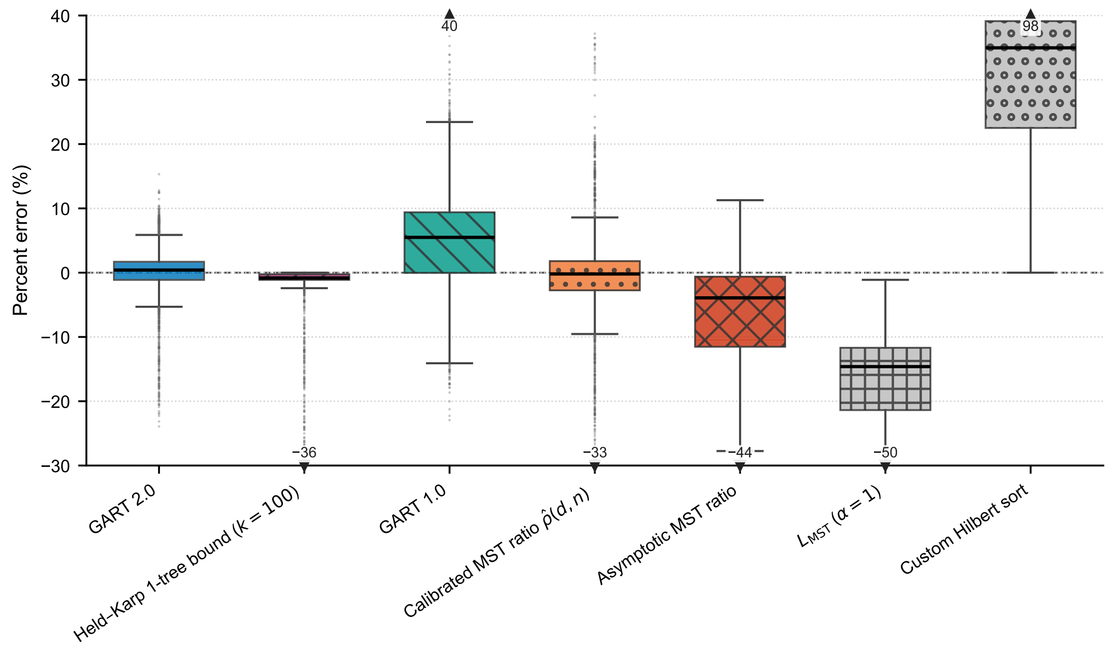

Figure 3: Percent error on the diverse 2D benchmark ($N = 2{,}580$ per estimator).

Table 4: 2D diverse benchmark results, by generator class.

		GART 2.0		GART 1.0		$\hat{\rho}(d,n)$		Asymptotic		$\alpha = 1$	
Bucket	Count	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE	MAPE	SDPE
Isotropic	840	**1.47**	**2.23**	5.38	5.18	3.31	5.11	7.12	8.77	17.1	7.79
Biased	840	**1.79**	**2.51**	7.93	7.97	3.21	5.24	8.22	10.1	18.4	8.97
Geometric: Grid†	210	**7.07**	**1.92**	11.9	4.97	16.0	6.42	9.43	6.76	4.54	6.01
Geometric: Boundary, X-Central	420	**1.96**	**2.60**	5.50	5.33	2.99	5.06	6.69	8.87	17.0	7.89
Clustered	60	**2.24**	**2.75**	5.41	6.71	7.94	5.36	10.5	6.36	20.5	5.65
Line Noise†	210	**10.8**	**6.09**	12.0	14.9	20.4	8.64	25.9	12.0	34.1	10.7
Total	2,580	**2.89**	**4.68**	7.30	7.90	5.76	9.38	9.20	11.6	18.0	10.3

† Not represented in training.

the middle on both. Both therefore confound unseen geometry with an extreme target value. On `grid` the $\alpha = 1$ floor is more accurate than the released model, the only row in this paper on which the floor beats it.

4.3.3 TSPLIB95 EUC_2D Benchmark

On the full EUC_2D benchmark GART 2.0 obtains 2.53% MAPE and 2.92% SDPE, against 3.53%/4.74% for the asymptotic ratio. This is the narrowest margin in the paper. α on these 78 instances has mean 1.13 and standard deviation 0.0558, so a single well-chosen constant is already close. The MAPE-minimizing constant on these instances is 1.13 and reaches 3.52%, so no constant multiplier does better on MAPE, and GART 2.0's 2.53% improves on that bound by 0.99 percentage points. The dispersion gap is the larger one, 2.92% SDPE against 4.74%, but SDPE alone cannot carry that comparison: for $\hat{L} = cL_{\mathrm{MST}}$ the signed error is $c/\alpha - 1$, so SDPE is proportional to c and shrinking c drives it to zero at ruinous MAPE. GART 2.0 attains the lower SDPE at the lower MAPE. Table 17 splits the 78 instances into three size buckets; the coarse $n > 400$ bucket should not be read as a homogeneous asymptotic regime.

In the coarse $n > 400$ bucket the GART 2.0 and asymptotic-ratio SDPE intervals overlap, on $[3.20, 3.62]$. In the post hoc $n > 10{,}000$ slice of five instances GART 2.0 is lower on both metrics, 0.57% MAPE and 0.72% SDPE against the asymptotic ratio's 2.74% and 1.46%. The largest instance, `d18512`, is more than $18\times$ the training-size cap. For points

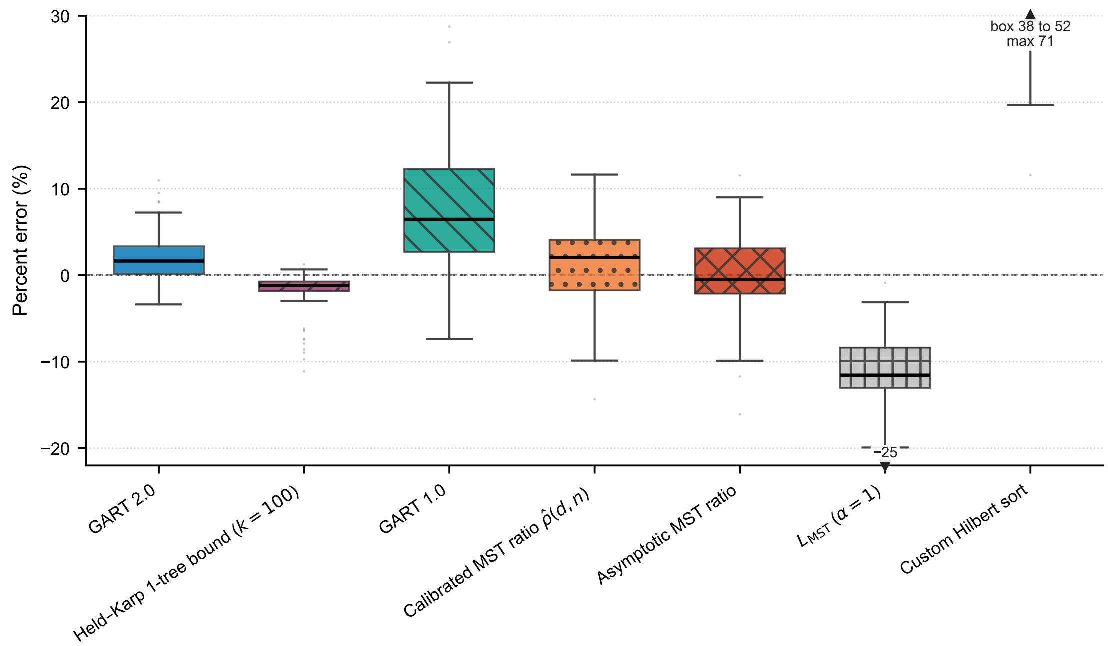

Figure 4: Percent error on the TSPLIB95 EUC_2D benchmark ($N = 78$ per estimator).

drawn independently from a bounded density, normalized TSP and MST lengths each converge to dimension-dependent constants [Beardwood et al., 1959, Steele, 1988], which motivates a fixed-ratio baseline but does not establish that the heterogeneous $n > 400$ bucket is asymptotic; the advantage at extreme sizes is an observation on five instances and not proof of a convergence mechanism.

Both classical estimators are far behind here. TSPLIB instances carry no uniform density on a known region, so each area-based estimator runs outside the conditions it was derived under. BHH receives a convex hull in place of a sampling region that does not exist. Çavdar–Sokol is evaluated on graphs without a node at the corners of their covering rectangle, a property every one of its training graphs had. Both overpredict. These rows measure the cost of that mismatch, not the quality of the cited methods.

4.4 Matched-Domain Comparison

Estimators marketed across domains do better on these generalized benchmarks than others.

BHH falls from 23.8% MAPE on the full 2D benchmark to 8.91% here. The region input and the instance set both change between its two panels, so that figure combines the two effects and does not isolate the region input. BHH's remaining error is a systematic underprediction, -8.65% signed against 7.76% SDPE, as an asymptotic constant evaluated at n as low as 5 should. Çavdar–Sokol falls from 18.2% to 8.16%; the estimator itself did not change between panels, so the drop is a domain effect. It has the lower MAPE of the two classical estimators on every corpus, and the lower median absolute error here, but not the lower dispersion. At 14.3% SDPE its dispersion is the widest in the panel, and its median absolute error is 2.84%, so it is close on most uniform instances and badly wrong on a few.

GART 2.0 beats both classical estimators on the matched domain, on both metrics. On the 210 uniform instances it obtains 1.31% MAPE against Çavdar–Sokol's 8.16%, BHH's 8.91% and the $\alpha = 1$ floor's 15.6%.

4.5 Generalization Beyond the Training Grid

Withholding both $d = 15$ and $d = 25$ from training, 11.8% of the training rows, raises test MAPE at $d = 15$ from 0.50% to 0.58% and at $d = 25$ from 0.35% to 0.38%, while the neighboring trained dimensions $d \in \{10, 20, 30\}$ do not degrade at all, moving from 0.43% to 0.42%. The loss is specific to the withheld strata rather than to the reduced training volume, and it takes the form of a bias shift: mean signed error at $d = 15$ moves from -0.008% to $+0.134\%$. Training only on $n \leq 200$ and testing on $n \in (200, 1000]$ raises MAPE from 0.41% to 0.59%, again almost entirely as a uniform offset, since SDPE moves only from 0.47% to 0.56% while mean signed error rises from $+0.28\%$ to $+0.51\%$, a factor of 1.8.

Three refits vary only the training data, holding hyperparameters and seed fixed; the baseline among them reproduces the released model bit-for-bit at 0.618% MAPE over the 16,846 scored rows. Off the grid it acquires a systematic offset and keeps its structure. We demonstrate no correction for that offset.

4.6 Failure on Near-Collinear Geometry

Section 4.3.2 reports 10.8% MAPE on the near-collinear Line Noise class against 1.47% on the isotropic class, and the error is almost entirely one-sided: MSPE is -10.8, so the model under-predicts.

TreeSHAP on the released booster, run over the 2,580 diverse 2D benchmark instances, shows why. On the Isotropic class (840 instances) the Greedy-to-MST Ratio carries 28.7% of the total mean $|\mathrm{SHAP}|$, the largest share of any feature; the MST Dominance Ratio is next at 19.3%, then dimension at 8.16%, n at 7.97%, and normalized MST diameter at 6.36%. Isotropic predictions are unbiased: mean predicted α is 1.22 against a mean true α of 1.22.

On Line Noise (210 instances) the same feature does not go silent and does not point the wrong way: its mean signed SHAP rises from $+0.417$ on Isotropic to $+0.732$ on Line Noise, and the MST Dominance Ratio's from $+0.105$ to $+0.369$, both correctly pushing α up. The prediction still falls short: mean predicted α is 1.38 against a mean true α of 1.56, a shortfall of 0.174, the largest of any class (the jittered-grid class is next at 0.0738). The training split itself contains zero instances with $\alpha > 1.5$ at $n \geq 200$; its α 99th percentile is 1.54, and 54% of the 210 Line Noise instances exceed it.

The reason is support, on exactly the feature carrying the prediction. On the training split the Greedy-to-MST Ratio spans $[1.06, 1.67]$ between its 1st and 99th percentiles; on Line Noise it averages 1.77, and 78.1% of the class's rows fall outside that band. The other top features stay inside it: the MST Dominance Ratio is 14.3% outside; dimension, n, and MST diameter are 0% outside. A boosted tree cannot predict past the highest split value it saw in training, so the output saturates while the true α keeps rising.

We tested two repairs against gates fixed before either candidate was scored, and rejected both. Extending the feature set moves the Line Noise slope at $n \geq 200$ from 0.294 to 0.569; adding training support alone moves it to 0.332; the two together reach 0.798, so the deficit is joint, not additive; the released model, not part of this ladder, already reaches 0.362 on the same slice. Augmenting the training split with matched instances cleared its gates, but half of the gain came from a $d = 2$ lattice generator identical to the benchmark's own `grid` generator, contaminating the result on the candidate's own geometry. A de-contaminated candidate cleared its gates too, but forfeited the paper's one distribution-independent dispersion claim: its advantage over the asymptotic ratio on TSPLIB fell to $p = 0.065$, against 0.0048 for the released model. We did not make that trade. A per-family multiplicative constant, an oracle a deployed estimator does not have, recovers 97.9% of the first candidate's gain by itself, so most of what either candidate buys is an offset, not a repaired response to geometry. The released model is unchanged.

4.7 Rank Agreement and Calibration

The estimator must *order* instances the way the stored references do even when its absolute numbers are off. The close-pair statistic below is a plain concordant fraction and sits at 0.5 under an unrelated ordering. Global rank correlation is close to 1 for every estimator here: cross-instance cost varies over orders of magnitude and L_{MST} already tracks that variation, so a high rank correlation demonstrates nothing about the learned ratio. Both metrics are reported with L_{MST} itself as the control. Every constant multiplier is a monotone transform of L_{MST} and scores identically to it, so the asymptotic-ratio and $\alpha = 1$ controls coincide on 2D and TSPLIB.

For pairs satisfying $|L_A - L_B| / \max(L_A, L_B) < 5\%$, GART 2.0 orders 74.5% of 74,826 2D pairs, 92.4% of 1,647,719 multidimensional pairs, and 72.2% of 54 TSPLIB pairs correctly, against 71.0%, 69.1%, and 61.1% for the $\alpha = 1$ control. The multidimensional gain of 23.4 percentage points is the substantive one; the 2D gain of 3.44 points is not, and on TSPLIB 54 pairs cannot support an 11.1-point claim. At a 10% threshold GART 2.0 obtains 83.6%, 96.1%, and 81.0%, against 77.8%, 75.9%, and 73.3% for the $\alpha = 1$ control. The percentages remain descriptive because the pairs share instances and no block-bootstrap interval is computed over them. GART 1.0 trails GART 2.0 on the TSPLIB close-pair statistic at the tighter threshold, at 70.4%.

The $n > 400$ TSPLIB bucket has $R^2_\alpha = -0.0710$: the model predicts α worse than the bucket mean does. That bucket has low target variance and an upward mean prediction offset while its tour-cost MAPE remains 2.90%, so the cost-level result does not remove the ratio-calibration failure. On TSPLIB EUC_2D the difference between GART 2.0 and the asymptotic MST ratio is -0.997 percentage points with a 95% interval of $[-1.77, -0.224]$ and $p = 0.0049$: on that benchmark GART 2.0's MAPE advantage is small but resolvable, and its larger gain is dispersion, 2.92% SDPE against 4.74%.

4.8 Computational Cost

Feature extraction dominates the estimator's cost. In the TSPLIB timing log GART 2.0 spends 92.9% of its wall time on feature extraction and MST construction against 5.03% on LightGBM inference. The Time column of Table 17 — ms, the median per-instance time in every table that follows — puts every estimator on one protocol: one estimator per process, a single thread, the median of 11 repeats. On that protocol GART 2.0 costs roughly twice the MST-ratio family and an order of magnitude over the two classical closed forms, which compute a convex hull and then evaluate an algebraic expression.

Those multiples are ratios of medians, not of summed totals, and are the wrong comparison for deployment, where the alternative is a solve. On the multidimensional benchmark the median prediction and the median reference-tour generation cost the same, 121 ms against 122 ms, since most of those instances are small enough to solve outright. The argument holds only where the solve is expensive: at $n \in [501, 1000]$ the reference tour takes 3,799 ms against GART 2.0's 460 ms, and on TSPLIB `d18512`, at $n = 18{,}512$ and more than $18\times$ the largest training instance, GART 2.0 predicts in 239 ms against the known lower bound, which costs 167 times as much at the crossing budget and 2,078 times as much, 493 seconds, at the top of the ladder.

5 Comparison Against a Known Lower Bound

Every baseline of Section 4.1 is an estimator. It returns a number with no guarantee, scored only by how close it lands, unlike the Held–Karp 1-tree bound. On the aggregate of all four benchmarks, and on TSPLIB EUC_2D once scaled by a single training-split constant, it is cheaper and more accurate than GART 2.0, so the estimator is off the cost/accuracy Pareto front of all four. However, the aggregates do not show how each family scales in n.

The certificate is one-sided. Because $\hat{w}_k$ never exceeds the optimum, its error against a reference tour is a duality gap plus whatever that reference sits above the optimum. The *calibrated* bound $c_k\,\hat{w}_k$ is also reported, where c_k is a single scalar fitted per budget on the planar training split and never on any scored instance. That row is not a certificate, since $c_k > 1$ pushes some predictions above the optimum, and it is not training-free. It is the like-for-like comparator against a trained estimator. Its constant is planar-fitted, so the multidimensional ladder reports the raw bound alone.

Both ascents run on all four benchmarks, and each headline table reports the arm that leads on its own corpus. The planar tables use the Volgenant–Jonker step schedule with Helsgaun's direction smoothing [Volgenant and Jonker, 1982, Helsgaun, 2000]. The multidimensional tables use a Polyak step [Polyak, 1969] against a constructive upper bound, a nearest-neighbor tour improved by 2-opt and built from coordinates alone. Neither ascent is proved to attain $\max_\pi w(\pi)$, so every accuracy figure below is a floor on what this family reaches.

5.1 Results on the Multidimensional Benchmark

At an ascent budget of 200 (Table 5) the bound reaches 0.111% MAPE against GART 2.0's 0.618%, a factor of 5.56, at 0.709 times the cost, and it wins the paired comparison on 82.9% of instances. It first overtakes GART 2.0 at a budget of 100, where it costs 0.400 times as much.

The bound wins every band except $d \in \{2, 3\}$, widest exactly where the estimator was never trained, and matching GART 2.0's accuracy there costs 1.27 times as much.

The domination is, however, an artifact of instance size. Of the 16,846 scored instances 10,569, or 62.7%, carry at most 100 nodes, a regime in which the relaxation is nearly exact and nearly free; a corpus mean over that composition is largely a statement about small instances rather than about the estimator. We draw 20 instances per (dimension, size band) cell, matched on size across dimensions so that the cost ratio reports a dimension effect and not a size effect, and price the ladder on them under the protocol of Appendix Table 14, where the bound's cost is far more load-sensitive than GART 2.0's, 1.25 against 1.05 between a quiet and a noisy window, so every published repeat is from one quiet window; accuracy in each cell is measured on every instance both methods score, between 108 and 2,209 of them. GART 2.0 is undominated except at small n and at $d = 100$, where the bound is both cheaper and more accurate. At $n \geq 200$ the estimator is on the cost/accuracy front at every dimension it was trained on, and it is dominated only at $d = 100$, the dimension withheld from training, where its own accuracy degrades to 0.824% against 0.0622% at $d = 50$.

In the $n \in [600, 1000]$ band the tightest rung, one budget row, we priced is both dearer and looser than GART 2.0's printed accuracy: $k = 500$ costs 0.784% at 246 ms at $d = 2$, $k = 200$ costs 0.450% at 144 ms at $d = 3$, and the same budget costs 0.291% at 86.2 ms at $d = 4$. The bound is $\Theta(kn^2)$ in every dimension, so the cost of matching a fixed accuracy grows quadratically while the estimator's cost carries no accuracy term, and outright reversal — GART 2.0 dominating — arrives only in the plane and at $d = 3$, from roughly 200 nodes upward.

The ascent never reads the stored label. Supplying the released optimum in place of the coordinates-only upper bound converges faster early and to the same value. On the 3,069 scored instances whose label the generation run recorded as a Concorde-proven optimum — a record of that run, not a certificate the released artifacts retain (Section 3.2) — the target is an optimum rather than a heuristic tour, and GART 2.0 scores 1.03% against the bound's 0.0458% at a budget of 200. And the relaxation closes exactly, returning the optimum rather than a bound, on 37.2% of the scored multidimensional split. Pairwise distances concentrate as d grows, so the integrality gap usually attributed to this relaxation is a planar fact.

Roughly one instance in seven exhausts the largest budget we ran, so at the top of the size range the reported accuracy is a floor. The cost column is therefore the weaker of the two, and a cell whose verdict turns on a few milliseconds should be read as a trade-off rather than as a ranking; budgets that close their gap terminate early, which is why a larger budget can occasionally read cheaper by a fraction of a millisecond. The separating constant favors the estimator at $n \geq 200$ and the bound below it, the quantity Table 6 measures; it is not a complexity argument and we do not extrapolate it past $n = 1000$, the largest size this corpus carries.

5.2 Complexity of the Estimator Families

The 1-tree cannot use the Delaunay shortcut the MST family relies on in the plane. The node potentials break the Euclidean metric, so after the first iteration the perturbed minimum spanning tree is not a subgraph of the triangulation. At 144 potential vectors drawn from real ascent trajectories the Delaunay-restricted 1-tree is heavier than the exact one at 141 of them: the bound sits a complexity class above the MST family, not a constant factor above it.

Table 5: Multidimensional cost/accuracy ladder, all 16,846 scored instances, Polyak ascent; conventions as in Appendix G.1. Only the raw bound is reported, the calibration constant being planar-fitted.

	$d \in \{2,3\}$, $N = 1{,}295$			$d \in [4,10]$, $N = 4{,}534$			$d \in [15,50]$, $N = 5{,}170$			$d = 100$, $N = 5{,}847$		
Row	ms	$\times$	MAPE	ms	$\times$	MAPE	ms	$\times$	MAPE	ms	$\times$	MAPE
GART 2.0	5.87	1.00	1.45	3.30	1.00	0.728	3.59	1.00	0.340	3.26	1.00	0.593
$k = 0$	2.43	0.235	12.2	0.127	0.0412	7.45	0.374	0.0740	3.83	0.234	0.0742	2.13
$k = 10$	3.51	0.315	7.39	0.237	0.0769	4.50	0.362	0.125	2.16	0.263	0.0827	1.03
$k = 25$	5.18	0.474	4.75	0.383	0.126	3.24	0.552	0.165	1.46	0.373	0.119	0.691
$k = 50$	7.37	0.690	3.18	0.620	0.207	1.97	0.918	0.274	0.870	0.494	0.154	**0.412**
$k = 100$	12.9	1.27	1.39	1.21	0.387	**0.692**	1.95	0.563	**0.301**	0.758	0.233	**0.147**
$k = 200$	23.0	2.18	0.557	2.15	0.685	**0.161**	3.30	0.953	**0.0534**	1.19	0.369	**0.0247**
$k = 500$	50.8	4.92	0.384	4.51	1.38	0.0674	6.05	1.75	0.0114	1.12	0.347	**0.00421**

Table 6: The multidimensional frontier stratified by size and dimension; "Dominates" means strictly better on cost and accuracy at the listed budgets. The bound is the Polyak arm.

d	n band	N	GART 2.0 MAPE (%)	GART 2.0 (ms)	Verdict against the ladder
2	[20, 100]	243	1.829	2.9	bound dominates ($k = 200,\ 500$)
	[200, 500]	108	0.663	5.7	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$)
	[600, 1,000]	135	0.429	10.0	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$)
3	[20, 100]	243	1.431	3.4	bound dominates ($k = 100,\ 200,\ 500$)
	[200, 500]	108	0.459	12.2	**GART 2.0 dominates** ($k = 100,\ 200$)
	[600, 1,000]	135	0.332	25.7	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200$)
4	[20, 100]	243	1.015	2.4	bound dominates ($k = 200$)
	[200, 500]	108	0.403	14.2	trade-off at every rung
	[600, 1,000]	135	0.251	58.2	**GART 2.0 dominates** ($k = 200$)
10	[20, 100]	243	0.566	2.4	bound dominates ($k = 200$)
	[200, 500]	108	0.204	14.7	trade-off at every rung
	[600, 1,000]	135	0.173	100.2	trade-off at every rung
50	[20, 100]	243	0.214	2.8	bound dominates ($k = 200$)
	[200, 500]	108	0.095	22.7	trade-off at every rung
	[600, 1,000]	134	0.062	114.3	trade-off at every rung
100	[20, 100]	2,209	0.501	3.4	bound dominates ($k = 100,\ 200$)
	[200, 500]	977	0.723	26.8	bound dominates ($k = 50,\ 100$)
	[600, 1,000]	1,185	0.824	120.4	bound dominates ($k = 50,\ 100$)

In the plane GART 2.0 is $\Theta(n \log n)$ and the bound is $\Theta(n^2 d + kn^2)$, and the measured log–log cost exponents in n agree: 0.975 for GART 2.0 above a thousand nodes against 2.08–2.13 for the bound over the same window. GART 2.0 costs a flat 1.99 to 2.34 times the MST-ratio family across four orders of magnitude in n; against the 1-tree at a fixed budget the ratio moves from 7.70 at a thousand nodes to 211 at sixteen thousand (Figure 5).

The separation is planar. On the multidimensional benchmark GART 2.0's own measured cost exponent in n is 2.02 at $d \in [4, 10]$ and 1.82 at $d \in [15, 50]$, against 1.78–1.89 for the bound, because the MST construction falls back to a dense all-pairs kernel from $d = 4$ upward. At $d \in \{2, 3\}$ that exponent is 0.556. On that benchmark both families are quadratic, and the separating constant runs against the estimator.

Branch-and-cut admits no polynomial bound, and the Held–Karp dynamic program [Held and Karp, 1962] is $\Theta(n^2 2^n)$.

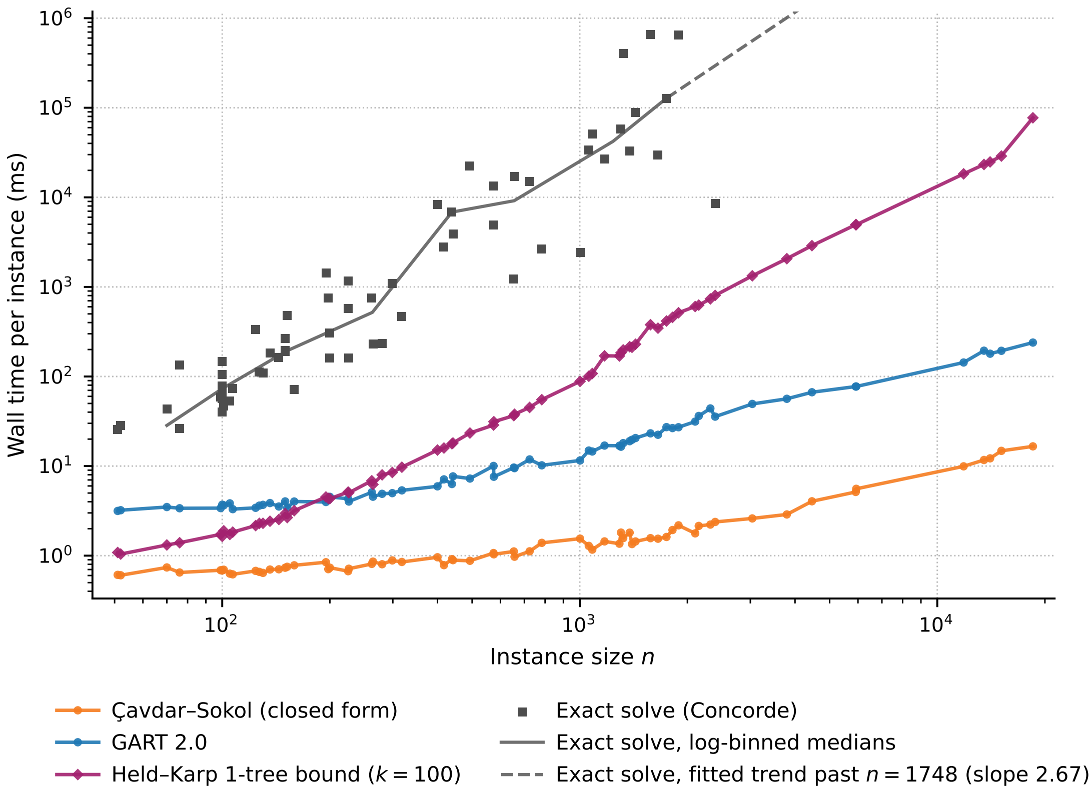

Figure 5: Per-instance cost against instance size on TSPLIB95 EUC_2D.

Table 7: Dominant per-instance cost of each family as the released code executes it.

Family	**Dominant term as implemented**	**Guarantee**	**Training**
Classical closed forms	$\Theta(d\,n\log n)$; a row-deduplication pass, not the arithmetic	None	Constants fitted in the source
Constant multiples of L_{MST}	$\Theta(n\log n)$ at $d = 2$ (Delaunay); $\Theta(n^2d + n^2\log n)$ otherwise	Lower bound only at $\alpha = 1$	The two calibrated rows only
Space-filling curve (Hilbert)	$\Theta(ndh)$ interpreted, plus an $O(n\log n)$ sort	Upper bound: it builds a tour	None
GART 2.0 (learned head on this feature block)	Feature cost as the MST family above, except the greedy tour's dense $\Theta(n^2)$ scan below its candidate-list threshold; head $O(1)$ in n	None	Yes
GART 1.0	$\Theta(n^2d)$ time and $\Theta(n^2)$ memory; a dense matrix for two scalars	None	Yes
Held–Karp 1-tree, budget k	$\Theta(n^2d + kn^2)$; no Delaunay path exists after the first iteration	Lower bound	None
Calibrated 1-tree	As above	None	One scalar per budget
Exact solver	$\Theta(n^2 2^n)$ for the Held–Karp dynamic program; branch-and-cut admits no polynomial bound	Exact	None

5.3 Operating Regime

GART 2.0 is cheaper than a 1-tree bound above roughly 400 nodes in the plane and dearer below roughly 150 (Appendix Table 14). Its accuracy sits between the bound's and the constant-ratio family's: against the MST-ratio family it is separated by 0.997 percentage points on the full 78-instance EUC_2D set, against 1.35 points from the converged bound on the same set.

On the aggregate of every benchmark here the bound is the better instrument, because the aggregate is dominated by small instances. Read at fixed size instead (Table 6), the ordering turns over: at $n \geq 200$ GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands it is strictly better than the bound on both axes from an ascent budget of 25 upward in the plane and over the mid-ladder budgets at $d = 3$.

Below roughly 200 nodes, prefer the certified bound at any dimension (Appendix Table 15): it is cheaper and more accurate, needs no training corpus, and tells the caller how wrong it can be. Above that size prefer the estimator at any dimension it was trained on, in the plane and at $d = 3$ where the ordering reverses, because each ascent iteration costs the bound $\Theta(n^2)$ at a fixed accuracy in every dimension while the estimator's planar cost stays subquadratic. At $d = 100$, outside the training grid, prefer the bound at every size: large n does not rescue the estimator there, and the reason is its own degradation off the grid, not the bound's strength. Given only a distance matrix, prefer the bound at any size (Appendix G.2). Where inputs are known to be independent, uniformly distributed on a known region and an error in the high single digits of percent is acceptable, the classical closed forms remain the cheapest instrument by an order of magnitude over GART 2.0, with no MST to build. Large instances re-evaluated many times under changing input are the last-mile and partitioning setting of Section 1. In that regime an exact solve is out of reach, a quadratic bound is already expensive, and GART 2.0's cost stays sub-quadratic.

6 Application: Non-Euclidean Estimation via MDS Embedding

TSPLIB95 contains 33 non-EUC_2D instances: 4 CEIL_2D, 2 ATT, 10 GEO, and 17 EXPLICIT. An exhaustive triangle-inequality scan identifies ten EXPLICIT matrices with structural violations (worst direct-edge ratio at least 1.2), which are outside every metric-dependent aggregate in this paper: `brg180`, `brazil58`, `gr120`, `gr48`, `gr24`, `gr21`, `bays29`, `hk48`, `dantzig42`, and `gr17`. That leaves 23 instances, seven of them EXPLICIT, and three of those (`swiss42`, `pa561`, and `fri26`) carry mild rounding-level violations, so conclusions are restricted to this screened set rather than to all non-Euclidean TSPs. We evaluate a hybrid pipeline that computes MST features from the original matrix and geometric features from a multidimensional-scaling (MDS) embedding. The paired reference is a constant multiple of the original-distance MST, a fixed $\alpha = 1.14$ close to the MAPE-minimizing constant 1.13 over the full 111-instance TSPLIB set. The MAPE-minimizing constant over these 23 instances is 1.17 and reaches 7.55%, which is the floor any constant multiplier faces on this metric.

6.1 MDS Preprocessing

CEIL_2D cases use their native coordinates. For ATT, GEO, and EXPLICIT cases, classical MDS gives an approximate Euclidean embedding of the distance matrix [Torgerson, 1952, Gower, 1966, Borg and Groenen, 2005, Cox and Cox, 2000]. We select the smallest embedding dimension κ retaining at least 99.9% of positive eigenvalue mass, subject to $\kappa \leq 100$, and pass κ as the model's `dimension` feature. The cap prevents five of 19 MDS cases from reaching 99.9%; hence this criterion does not imply distance preservation.

6.2 Results on Non-Euclidean TSPLIB95

On the instances it accepts, the hybrid pipeline clears the 7.55% constant-multiplier floor.

Table 8: Paired evaluation on the 23 screened non-EUC_2D instances. GART 2.0 declines `si1032`, so the Total row is not like-for-like.

Distance type	Count	Estimator	N	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)
ATT	2	GART 2.0	2	1.42 [—, —]	3.66	3.66	−3.66
		Fixed $\alpha = 1.136$	2	4.12 [—, —]	3.38	3.38	−3.38
CEIL_2D	4	GART 2.0	4	5.16 [0.112, 6.35]	5.77	6.05	4.04
		Fixed $\alpha = 1.136$	4	6.44 [1.08, 7.86]	7.54	7.77	5.95
GEO	10	GART 2.0	10	2.60 [1.48, 3.23]	2.51	2.90	−1.63
		Fixed $\alpha = 1.136$	10	9.49 [2.13, 10.9]	9.62	4.10	−9.62
EXPLICIT screened	7	GART 2.0	6	3.11 [1.44, 3.87]	3.01	2.35	−2.15
		Fixed $\alpha = 1.136$	7	9.55 [5.04, 10.9]	8.12	10.2	1.75
Total	**23**	**GART 2.0**	**22**	**3.89 [2.52, 4.79]**	**3.34**	**3.21**	**−0.925**
		Fixed $\alpha = 1.136$	23	10.5 [6.94, 12.9]	8.26	6.29	−2.91

GART 2.0 declines `si1032` rather than extrapolate below the greedy-to-MST ratio it was trained on. α ranges from 1.01 to 1.51 across the set, and a single multiplier cannot track that spread without trading one metric against the other. On the two ATT instances the fixed constant has the lower MAPE, and on CEIL_2D the two estimators are close in SDPE. Groups of two and four instances cannot separate estimators. We do not estimate the relationship between per-instance embedding distortion and prediction error, so this remains a feasibility study for the hybrid pipeline rather than evidence of embedding robustness.

The Held–Karp relaxation never invokes the triangle inequality, so it is defined on any symmetric matrix and reads these instances directly. On the 29 instances both methods score over all 33 non-EUC_2D files — a set wider than the metric screen above, since the certificate tolerates non-metric input — GART 2.0 obtains 3.81% MAPE against 1.27% for the raw bound at an ascent budget of 100 under the Volgenant–Jonker step and 0.509% at a budget of 500 under the Polyak step, a factor of 7.50. The bound also scores `brg180` and `si1032`.

7 Conclusion

The GART 2.0 framework, trained on synthetic data, is competitive on real-world problems. Exposing the model to more relevant training data can move it from a prototype to a viable product.

It is more accurate on the multidimensional benchmark at lower cost, and more accurate again on the non-Euclidean instances it reads directly and GART 2.0 reaches only through an embedding. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on. The estimator's advantage is scale. In the plane the measured cost exponents put the two families in different complexity classes (Section 5), and an instance many times larger than anything in training is predicted in a fraction of a second. Below that size, off the training grid in dimension, or given only a distance matrix, the known lower bound should be preferred. The model weights, the scored benchmark outputs, and the scripts that regenerate every table and figure are released (Appendix A).

Accuracy depends on geometry far more than on scale. Error on the near-collinear class is several times what it is on the isotropic class. Section 4.6 diagnoses that failure as a training-coverage gap and a feature gap at once. Two pre-registered repair attempts were rejected and the released model is unchanged. Withholding a whole dimension or the large-n range from training costs a fraction of a MAPE point and appears as a systematic offset rather than as scatter. The estimator therefore degrades predictably off its training grid, and we demonstrate no correction for that offset. The labels are not uniformly certificates. Some are proven optimal by a 1-tree bound that closes on the tour, some by exact dynamic programming, and the remainder carry an upper bound whose residual on the scored test split is narrow but not zero.

Can a learned estimator understand the relationship between the MST and the TSP to create an efficient tour-length estimator?

References

R. Agarwala, D. L. Applegate, D. Maglott, G. D. Schuler, and A. A. Schäffer. A fast and scalable radiation hybrid map construction and integration strategy. *Genome Research*, 10(3):350–364, 2000.

T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 2623–2631. ACM, 2019. doi: 10.1145/3292500.3330701.

D. L. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006.

J. J. Bartholdi, III and L. K. Platzman. An O(N log N) planar travelling salesman heuristic based on spacefilling curves. *Operations Research Letters*, 1(4):121–125, 1982. doi: 10.1016/0167-6377(82)90012-8.

J. Beardwood, J. H. Halton, and J. M. Hammersley. The shortest path through many points. *Mathematical Proceedings of the Cambridge Philosophical Society*, 55(4):299–327, 1959. doi: 10.1017/S0305004100034095.

D. J. Bertsimas and G. van Ryzin. An asymptotic determination of the minimum spanning tree and minimum matching constants in geometrical probability. *Operations Research Letters*, 9(4):223–231, 1990. doi: 10.1016/0167-6377(90)90066-E.

A. Bircher, M. Kamel, K. Alexis, M. Burri, P. Oettershagen, S. Omari, T. Mantel, and R. Siegwart. Three-dimensional coverage path planning via viewpoint resampling and tour optimization for aerial robots. *Autonomous Robots*, 40(6):1059–1078, 2016.

I. Borg and P. J. Groenen. *Modern Multidimensional Scaling: Theory and Applications*. Springer, 2nd edition, 2005. doi: 10.1007/0-387-28981-X.

J. Bossek, P. Kerschke, A. Neumann, M. Wagner, F. Neumann, and H. Trautmann. Evolving diverse TSP instances by means of novel and creative mutation operators. In *Proc. 15th ACM/SIGEVO Conference on Foundations of Genetic Algorithms (FOGA XV)*, pages 58–71, 2019. doi: 10.1145/3299904.3340307.

J. Bourgain. On Lipschitz embedding of finite metric spaces in Hilbert space. *Israel Journal of Mathematics*, 52(1–2):46–52, 1985. doi: 10.1007/BF02776078.

B. Çavdar. *A computation-implementation parallelization approach to time-sensitive applications*. PhD thesis, Georgia Institute of Technology, Atlanta, GA, 2014. URL `https://repository.gatech.edu/handle/1853/52322`. H. Milton Stewart School of Industrial and Systems Engineering; advisor Joel S. Sokol. Chapter 4 is the tour-length estimation model later published as Çavdar and Sokol (2015).

B. Çavdar and J. Sokol. A distribution-free TSP tour length estimation model for random graphs. *European Journal of Operational Research*, 243(2):588–598, 2015. doi: 10.1016/j.ejor.2014.12.020.

B. Chazelle. An optimal convex hull algorithm in any fixed dimension. *Discrete & Computational Geometry*, 10(4):377–409, 1993. doi: 10.1007/BF02573985.

T. W. Chien. Operational estimators for the length of a traveling salesman tour. *Computers & Operations Research*, 19(6):469–478, 1992. doi: 10.1016/0305-0548(92)90002-M.

Y. Choi. *Adjustment Factors and Applications for Analytic Approximations of Tour Lengths*. PhD thesis, University of Maryland, College Park, 2021. DRUM handle 1903/27698.

N. Christofides. Worst-case analysis of a new heuristic for the travelling salesman problem. Technical Report 388, Carnegie-Mellon University, 1976. Reprinted in Operations Research Forum 3:20, 2022, doi:10.1007/s43069-021-00101-z.

W. J. Cook and A. Rohe. Computing minimum-weight perfect matchings. *INFORMS Journal on Computing*, 11(2):138–148, 1999.

M. Cortina-Borja and T. Robinson. Estimating the asymptotic constants of the total length of Euclidean minimum spanning trees with power-weighted edges. *Statistics & Probability Letters*, 47(2):125–128, 2000. doi: 10.1016/S0167-7152(99)00147-9.

T. F. Cox and M. A. Cox. *Multidimensional Scaling*. Chapman and Hall/CRC, 2nd edition, 2000. ISBN 9781584880943.

C. F. Daganzo. The distance traveled to visit N points with a maximum of C stops per vehicle: An analytic model and an application. *Transportation Science*, 18(4):331–350, 1984a. doi: 10.1287/trsc.18.4.331.

C. F. Daganzo. The length of tours in zones of different shapes. *Transportation Research Part B: Methodological*, 18(2):135–145, 1984b. doi: 10.1016/0191-2615(84)90027-4.

G. B. Dantzig, D. R. Fulkerson, and S. M. Johnson. Solution of a large-scale traveling-salesman problem. *Operations Research*, 2(4): 393–410, 1954. doi: 10.1287/opre.2.4.393.

B. Delaunay. Sur la sphère vide. *Bulletin de l'Académie des Sciences de l'URSS, Classe des Sciences Mathématiques et Naturelles*, 6:793–800, 1934.

J. Edmonds. Paths, trees, and flowers. *Canadian Journal of Mathematics*, 17:449–467, 1965. doi: 10.4153/CJM-1965-045-4.

S. Feldman. Tour length estimation guided vehicle routing. Master's thesis, Rensselaer Polytechnic Institute, 2025. URL `https://search.proquest.com/openview/d4a8aaeb70d19bb62af72359b5caf7f7/1?pq-origsite=gscholar&cbl=18750&diss=y`. Introduces the original GART 1.0 tour-length estimator. Available via ProQuest Dissertations Publishing.

M. A. Figliozzi. Planning approximations to the average length of vehicle routing problems with time window constraints. *Transportation Research Part B: Methodological*, 43(4):438–447, 2009. doi: 10.1016/j.trb.2008.08.004.

Finmile. Revolutionizing last-mile delivery for unparalleled efficiency and customer satisfaction. Finmile industry resource page, 2025. URL `https://finmile.co/resources/finmile-revolutionizing-last-mile-delivery`. Accessed 2026-08-26.

H. Freeman and R. Shapira. Determining the minimum-area encasing rectangle for an arbitrary closed curve. *Communications of the ACM*, 18(7):409–413, 1975. doi: 10.1145/360881.360919.

J. H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

H. N. Gabow. Data structures for weighted matching and nearest common ancestors with linking. In *Proceedings of the 1st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 434–443, 1990.

B. L. Golden and W. R. Stewart. Empirical analysis of heuristics. In E. L. Lawler, J. K. Lenstra, A. H. G. Rinnooy Kan, and D. B. Shmoys, editors, *The Traveling Salesman Problem: A Guided Tour of Combinatorial Optimization*, pages 207–249. Wiley, 1985.

J. C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3–4):325–338, 1966. doi: 10.1093/biomet/53.3-4.325.

C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant. Array programming with NumPy. *Nature*, 585: 357–362, 2020. doi: 10.1038/s41586-020-2649-2.

M. Held and R. M. Karp. A dynamic programming approach to sequencing problems. *Journal of the Society for Industrial and Applied Mathematics*, 10(1):196–210, 1962. doi: 10.1137/0110015.

M. Held and R. M. Karp. The traveling-salesman problem and minimum spanning trees. *Operations Research*, 18(6):1138–1162, 1970. doi: 10.1287/opre.18.6.1138.

M. Held and R. M. Karp. The traveling-salesman problem and minimum spanning trees: Part II. *Mathematical Programming*, 1(1): 6–25, 1971. doi: 10.1007/BF01584070.

K. Helsgaun. An effective implementation of the Lin-Kernighan traveling salesman heuristic. *European Journal of Operational Research*, 126(1):106–130, 2000.

K. Helsgaun. An extension of the Lin-Kernighan-Helsgaun TSP solver for constrained traveling salesman and vehicle routing problems. Technical report, Roskilde University, 2017.

D. Hilbert. Ueber die stetige Abbildung einer Linie auf ein Flächenstück. *Mathematische Annalen*, 38(3):459–460, 1891. doi: 10.1007/BF01199431.

C. Holland, J. Levis, R. Nuggehalli, B. Santilli, and J. Winters. UPS optimizes delivery routes. *Interfaces*, 47(1):8–23, 2017. doi: 10.1287/inte.2016.0875. Winner of the 2016 Franz Edelman Award.

D. S. Johnson, L. A. McGeoch, and E. E. Rothberg. Asymptotic experimental analysis for the Held–Karp traveling salesman bound. In *Proceedings of the 7th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 341–350, 1996.

I. Jolliffe. *Principal Component Analysis*. Springer, 2002.

A. R. Karlin, N. Klein, and S. Oveis Gharan. A (slightly) improved approximation algorithm for metric TSP. In *Proceedings of the 53rd Annual ACM Symposium on Theory of Computing*, pages 32–45, 2021. doi: 10.1145/3406325.3451009.

R. M. Karp. Reducibility among combinatorial problems. In R. E. Miller and J. W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972. doi: 10.1007/978-1-4684-2001-2_9.

G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.

S. Kou, B. L. Golden, and S. Poikonen. Optimal TSP tour length estimation using standard deviation as a predictor. *Computers & Operations Research*, 148:105993, 2022. doi: 10.1016/j.cor.2022.105993.

S. Kou, B. L. Golden, and S. Poikonen. Optimal TSP tour length estimation using Sammon maps. *Optimization Letters*, 17(1): 89–105, 2023. doi: 10.1007/s11590-022-01937-y.

S. Kou, B. L. Golden, and S. Poikonen. Estimating optimal objective values for the TSP, VRP, and other combinatorial problems using randomization. *International Transactions in Operational Research*, 31(5):3443–3458, 2024. doi: 10.1111/itor.13260.

O. Kwon, B. L. Golden, and E. A. Wasil. Estimating the length of the optimal TSP tour: An empirical study using regression and neural networks. *Computers & Operations Research*, 22(10):1039–1046, 1995. doi: 10.1016/0305-0548(94)00093-N.

S. K. Lam, A. Pitrou, and S. Seibert. Numba: A LLVM-based Python JIT compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015. doi: 10.1145/2833157.2833162.

S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017.

R. Mariescu-Istodor and P. Fränti. Solving the large-scale TSP problem in 1 h: Santa Claus Challenge 2020. *Frontiers in Robotics and AI*, 8:689908, 2021. doi: 10.3389/frobt.2021.689908.

X. Pan, Y. Jin, Y. Ding, M. Feng, L. Zhao, L. Song, and J. Bian. H-TSP: Hierarchically solving the large-scale traveling salesman problem. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 9345–9353, Washington, DC, USA, 2023. AAAI Press. doi: 10.1609/aaai.v37i8.26120.

C. H. Papadimitriou. The Euclidean traveling salesman problem is NP-complete. *Theoretical Computer Science*, 4(3):237–244, 1977. doi: 10.1016/0304-3975(77)90012-3.

F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

A. G. Percus and O. C. Martin. Finite size and dimensional dependence in the Euclidean traveling salesman problem. *Physical Review Letters*, 76(8):1188–1191, 1996. doi: 10.1103/PhysRevLett.76.1188.

B. T. Polyak. Minimization of unsmooth functionals. *USSR Computational Mathematics and Mathematical Physics*, 9(3):14–29, 1969. doi: 10.1016/0041-5553(69)90061-5.

R. C. Prim. Shortest connection networks and some generalizations. *Bell System Technical Journal*, 36(6):1389–1401, 1957.

G. Reinelt. TSPLIB—a traveling salesman problem library. *ORSA Journal on Computing*, 3(4):376–384, 1991. doi: 10.1287/ijoc.3.4.376.

D. J. Rosenkrantz, R. E. Stearns, and P. M. Lewis, II. An analysis of several heuristics for the traveling salesman problem. *SIAM Journal on Computing*, 6(3):563–581, 1977. doi: 10.1137/0206041.

A. I. Serdyukov. On some extremal tours in graphs (in Russian). *Upravlyaemye Sistemy*, 17:76–79, 1978.

K. Smith-Miles, J. I. van Hemert, and X. Y. Lim. Understanding TSP difficulty by learning from evolved instances. In *Learning and Intelligent Optimization*, volume 6073 of *Lecture Notes in Computer Science*, pages 266–280. Springer, 2010. doi: 10.1007/978-3-642-13800-3_29.

J. M. Steele. Growth rates of Euclidean minimal spanning trees with power weighted edges. *The Annals of Probability*, 16(4): 1767–1787, 1988. doi: 10.1214/aop/1176991596.

W. S. Torgerson. Multidimensional scaling: I. Theory and method. *Psychometrika*, 17(4):401–419, 1952. doi: 10.1007/BF02288916.

T. Varol, O. Ö. Özener, and E. Albey. Neural network estimators for optimal tour lengths of traveling salesperson problem instances with arbitrary node distributions. *Transportation Science*, 58(1):45–66, 2024. doi: 10.1287/trsc.2022.0015.

P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

T. Volgenant and R. Jonker. A branch and bound algorithm for the symmetric traveling salesman problem based on the 1-tree relaxation. *European Journal of Operational Research*, 9(1):83–89, 1982. doi: 10.1016/0377-2217(82)90015-7.

Y. Yuan and Y. Peng. RaceTrack: An approximation algorithm for the mobile sink routing problem. In *Ad-Hoc, Mobile and Wireless Networks*, volume 6288 of *Lecture Notes in Computer Science*, pages 135–148. Springer, 2010. doi: 10.1007/978-3-642-14785-2_11.

A Code and Data Availability

Our implementation, trained model weights, and benchmark scripts are available at `https://github.com/sfeldmanMIG25/Area-and-Distribution-Free-Estimator-for-TSP`. We release the per-class, per-dimension, per-size, rank-agreement, and paired-test grids as tidy tables with the per-instance predictions, and the provided scripts regenerate the synthetic instances. The classical estimators and the calibrated constant-ratio baselines are released as separate modules; the fitted constant-ratio table is frozen as a released artifact, not refitted at inference. A verification pass re-derives all 425 benchmark table cells against the typeset value, and a second pass does the same for the 182 cells of the cost/accuracy ladders of Section 5. The remaining tables, and scalars quoted in the running prose such as the timing split and the quarantine and bound-check counts, are transcribed by hand from the same artifacts and are not covered by either check. Label validation (Appendix E), the refits of Section 4.5, the MDS diagnostics (Appendix J), and the figures each have a standalone reproduction script. For direct audit of target clipping, the two released out-of-interval rows are `N1000_D15_G100_pchergphhtgncoe_59` (training, raw $\alpha = 0.975$) and `N5_D2_G100_oc_33` (validation, raw $\alpha = 2.06$).

The GART 2.0 pipeline uses SciPy [Virtanen et al., 2020] for Delaunay triangulation [Delaunay, 1934] and MST construction, Numba [Lam et al., 2015] for centroid-distance computation, NumPy [Harris et al., 2020] symmetric eigendecomposition for classical MDS, LightGBM [Ke et al., 2017] for regression, and scikit-learn [Pedregosa et al., 2011] for metric computation. Estimator timings were collected on one Intel i7 13th Gen machine with 64 GB RAM and Windows 11. The cold protocol times one harness invocation per instance, interpreter and just-in-time warm-up included; the protocol of Section 4.8 runs one estimator per process, warm-up completed before the clock starts. Concorde 03.12.19 and LKH-3 3.0.13 generated the stored reference tour costs; the canonical table does not retain solver certificates.

B Training Data Generation

Synthetic TSP instances span four distribution classes, with the generator assigned independently per axis.

B.1 Training Corpus

The uniform class draws coordinates from $\mathcal{U}(0, G)$ where G is the grid size. The normal class draws from $\mathcal{N}(G/2, G/6)$ truncated to $[0, G]$. The clustered class places $k \in [3, 15]$ cluster centers uniformly and draws points from $\mathcal{N}(\mu_k, \sigma_k)$ around each center. The correlated class uses autoregressive structure across axes, $x_{j+1} = \rho \cdot x_j + \epsilon$ with j indexing the axis, producing axis-aligned anisotropy.

Node counts span $n \in [5, 1{,}000]$. Training-visible dimensions run from 2 to 10 in steps of one and from 15 to 50 in steps of five, plus a held-out bucket at $d = 100$ (5,904 test instances). The pipeline ran Concorde and LKH-3 and retained the shorter tour cost.

The 106,272 solved instances are partitioned into training (69,768, 65.7%), validation (19,584, 18.4%), and test (16,920, 15.9%) using a stratified sample on $(d, n, \text{grid size})$. The stratification rule ranks instances within each stratum by a random uniform draw and assigns roughly the first tenth to test and the next fifth to validation, with the remainder to training. $d = 100$ instances bypass this rule and are locked to the test partition, which lifts the corpus-level test share to the 15.9% above. The per-(d, n) training-instance counts are released with the artifacts; grid size $G \in \{100, 1{,}000, 10{,}000\}$ is sampled uniformly within each cell.

B.2 2D Benchmark Generator Design

Four classes (isotropic, biased, geometric, clustered) follow conventions from the TSP-instance-generation literature, including the variants Çavdar and Sokol [2015] reports for the distribution-free baseline. The fifth class, Line Noise,

Table 9: Multidimensional synthetic benchmark grid: instance count per (d, n) bucket.

$d \setminus n$	$[5, 10]$	$[11, 50]$	$[51, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$	Total
$d = 2$	162	108	135	27	81	135	**648**
$d \in [3, 5]$	486	324	405	81	243	405	**1,944**
$d \in [6, 10]$	810	540	675	135	405	675	**3,240**
$d \in [15, 25]$	486	324	405	81	243	405	**1,944**
$d \in [30, 50]$	810	540	675	135	405	675	**3,240**
$d = 100$	1,476	984	1,230	246	738	1,230	**5,904**
Total	**4,230**	**2,820**	**3,525**	**705**	**2,115**	**3,525**	**16,920**

samples $y = mx + b + \epsilon$ with small ϵ and clips out-of-range coordinates to the grid boundary instead of resampling them. Its slope reaches ± 5 while x is drawn across the full grid width, so most points fall outside the box and are pinned to a face: the realized geometry is a union of two to four thin segments with dense collinear masses along the boundary, not a single noisy line. At $n \geq 200$ a median 59.9% of points lie exactly on a face, and α tracks that fraction closely (Spearman 0.86, rising monotonically from 1.18 to 1.43 across its quartiles) with no relationship to the transverse noise width. The `grid` member of the Geometric Struct. class draws a jittered square lattice, a pattern the four independent per-axis draws of Section 3.2 cannot produce; the training corpus contains no lattice instance.

B.3 Multidimensional Benchmark Grid

Table 9 gives the instance count in every (d, n) bucket of the 16,920-instance multidimensional benchmark.

C Feature Specifications

Table 10 lists the 11 geometric and centroid features, Tables 11 and 12 the 19 MST-derived ones, and Table 13 the single constructive ratio, 31 in all (Section 3.1). The MST edge moments are candidate predictors of the α ratio, not a tested causal mechanism. We introduce the Dominance, Gap, and Normalized Diameter ratios as $O(n)$ candidate proxies for cluster, filament, and hub structure, and the Leaf Ratio rests on the hypothesis that low-leaf trees are more common on linear manifolds than in isotropic clouds. This interpretation is not evaluated independently here. The Greedy-to-MST Ratio ranks second by SHAP magnitude (Appendix D) and is the feature the non-Euclidean coverage gate of Section 6 tests. Listed complexities assume the MST has already been built; MST construction itself dominates the asymptotic cost and is analyzed in Section 3.

D SHAP Feature Importance

We report the SHAP values [Lundberg and Lee, 2017] against the released booster on a 5,000-row sample of the 16,920-row held-out test split. The complete 31-feature ranking is released with the artifacts.

By group, the nineteen MST-derived features carry 50.7% of total mean absolute SHAP magnitude, the eleven geometric features 25.5%, and the single constructive ratio 23.9%. SHAP magnitude describes model reliance, not a causal effect or proof that a feature group improves generalization.

SHAP analysis served as a diagnostic during feature reduction and yielded the 31-feature set described in Section 3.1. The selection used validation feedback, so it is not an independent confirmatory test.

E Label Validation and Verification

Every label is screened before any table is built. We recompute the length of every stored tour on its own released coordinates, over all 106,272 instances. A stored cost that its own tour cannot reproduce to within rounding describes

Table 10: Geometric and centroid feature specifications.

Feature Name	Description & Rationale	Complexity	Origin / Citation
Dimension (d)	The number of spatial coordinates defining each point.	$O(1)$	Standard
Number of Nodes (n)	The total count of nodes in the instance; released feature name `n_customers`.	$O(1)$	Standard
Aspect Ratio	Ratio of the largest dimension range to the smallest. Detects subspace distortion.	$O(n \cdot d)$	**Proposed**
Bounding Hypervolume	Product of coordinate ranges: $\prod_{j=1}^{d}(\max_j - \min_j)$. Approximates the sampling-region measure $\mathcal{V}$ for BHH bounds; stored capped at e^{690}, beyond which the log twin carries the scale.	$O(n \cdot d)$	Adapted from Beardwood et al. [1959]
Log Bounding Hypervolume	Logarithm of hypervolume: $\sum_{j=1}^{d} \ln(\max_j - \min_j)$. Log-space stable for $d \geq 100$.	$O(n \cdot d)$	**Proposed**
Node Density	Ratio of n to Bounding Hypervolume. Proxy for point intensity $\delta = n/\mathcal{V}$.	$O(1)$	**Proposed**; cf. Beardwood et al. [1959]
Log Node Density	Logarithm of node density: $\ln(n) - \sum \ln(\max_d - \min_d)$. Log-space stable.	$O(1)$	**Proposed**
Centroid Dist (Mean, Std)	Average and deviation of distances to the centroid. Measures central tendency.	$O(n \cdot d)$	Adapted from Çavdar and Sokol [2015]
Centroid Dist Max	Radius of the enclosing sphere.	$O(n \cdot d)$	Adapted from Çavdar and Sokol [2015]
Centroid Dist IQR	Interquartile range of centroid distances. Robust dispersion metric.	$O(n \cdot d)$	**Proposed**; extends Çavdar and Sokol [2015]

Table 11: MST edge distribution statistics (10 features).

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Edge Max	Weight of the longest edge in the MST.	$O(n)$	Standard
MST Quantiles	10%, 25%, 50%, 75%, 90% percentiles. Provides a non-parametric distribution profile.	$O(n \log n)$	**Proposed**; via MST [Prim, 1957]
MST Edge Mean	Average edge length in the MST.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Std	Standard deviation of edge lengths in the MST.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Skewness	Asymmetry of edge distribution. High positive skew implies many short edges and few long bridges.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Kurtosis	Tailedness of distribution. Indicates the presence of extreme outliers.	$O(n)$	Smith-Miles et al. [2010]

Table 12: Topological and clustering feature specifications (9 features).

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Dominance Ratio	Sum of largest $\sqrt{n}$ edges / Total Length. Captures weight of inter-cluster bridges.	$O(n)$	**Proposed**; via MST [Prim, 1957]
MST Gap Ratio	Max edge / Median edge. Measures magnitude of sparsity jumps.	$O(1)$	**Proposed**; via MST [Prim, 1957]
MST Leaf Ratio	Fraction of nodes with degree $k = 1$. Topology proxy for dimension.	$O(n)$	Smith-Miles et al. [2010]
MST Degree (Mean, Std)	Branching factor statistics.	$O(n)$	Smith-Miles et al. [2010]
MST Degree Max	Maximum node degree. Identifies "hub" nodes common in High-D spaces.	$O(n)$	Smith-Miles et al. [2010]
MST Diameter	Sum of edge weights on the longest path between any two leaves, computed via double-BFS.	$O(n)$	Graph Theory
MST Normalized Diameter	Diameter / Total Length. Scale-invariant shape proxy ($= 1.0$ for path-manifolds exactly).	$O(1)$	**Proposed**; via double-BFS
Large Edge Count	Count of edges $> \mu + \sigma$. Secondary cluster indicator.	$O(n)$	**Proposed**; via MST [Prim, 1957]

Table 13: Constructive ratio specification.

Feature Name	Description & Rationale	Complexity	Origin / Citation
Greedy-to-MST Ratio	Length of an exact greedy nearest-neighbor tour divided by L_{MST}. A greedy tour is feasible, so the ratio is a constructed upper bound on the target α. Values outside $[1.035, 2.209]$, the range spanned by the full 106,272-row corpus, trigger the coverage gate; the training split alone spans $[1.0465, 2.1295]$.	$O(n^2 d)$ dense for $n \leq 3000$; $O(nmd)$ with $m = 16$ candidate neighbors above it	**Proposed**; greedy nearest-neighbor construction

no released point set, and no bound is tight enough to reconstruct the optimum from the coordinates alone. The 184 instances (0.173%) that fail are *quarantined*, not scored: they carry an empty reference cost and the status `quarantined_label_unrecoverable` in every released results CSV, and the scoring harness raises on any attempt to score one. An aggregate that ignores the status column yields NaN, never a plausible number. 74 of the 184 fall in the multidimensional test partition, so it is scored on the remaining $16{,}846$ instances throughout.

If B is a proven lower bound on an instance's optimum and L is its stored label, $B \leq \mathrm{OPT} \leq L$ must hold whenever L is an upper bound on the optimum. $B > L$ refutes the label outright, and no such tour enters the test. Every reference cost is recomputed in float64 from the released coordinates. At $n \leq 10$ the label is the exact Held–Karp dynamic-programming optimum, a certificate. Above that size the label is the float64 length of the retained tour, an upper bound that is exact wherever that tour is optimal; a 1-tree bound closes on the tour and proves it optimal after the fact for $2{,}182$ instances. The 2D benchmark's labels are the better of the stored tour and a float-metric 2-opt descent from it, because a solver that minimizes a rounded metric can select a tour the float64 metric beats. TSPLIB is scored on its published optima, with one documented exception: `linhp318` declares a fixed-edge Hamiltonian-path problem on `lin318`'s coordinates, so its published 41,345 is a path optimum no tour on those coordinates can attain. Every estimator and the bound are scored there against the tour optimum on those coordinates, $42{,}029$, `lin318`'s published value. A scan of the full instance set found no second fixed-edge declaration and no second duplicated geometry.

Across the 145,013 bound checks the released artifacts supply there are zero violations of $B \leq L$, the worst relative excess being 5.8×10^{-15}. The 184 quarantined instances above carry no recoverable label and are outside this check. Seven TSPLIB instances carry no bound: the test runs in TSPLIB's integer metric there, and the dense perturbed matrix does not fit in memory. Five of the seven are inside the scored EUC_2D set. Nothing in this section is claimed about those instances.

F Benchmark-Model Implementation

We attribute the tour-length estimators of Daganzo [Daganzo, 1984a], Chien [Chien, 1992], and Kwon–Golden–Wasil [Kwon et al., 1995], surveyed in Figliozzi [2009]. They belong to the same class of estimator, and we score BHH and Çavdar–Sokol as its representatives.

F.1 BHH

BHH assumes points drawn independently from a density f of bounded support: $L_n/n^{(d-1)/d} \to \beta_d \int f(x)^{(d-1)/d}\,dx$ [Beardwood et al., 1959]. For f uniform on a region of measure $\mathcal{V}$ the integral reduces to $\mathcal{V}^{1/d}$, giving $\beta_d n^{(d-1)/d}\mathcal{V}^{1/d}$. The convex hull of a realized sample is a different, downward-biased statistic that converges to $\mathcal{V}$ only as n grows. Our synthetic generators draw on $[0, G]^d$, so $\mathcal{V} = G^d$ is exact, and we report that form on the matched domain. We use $\beta_2 = 0.7124 \pm 0.0002$ [Johnson et al., 1996] and $\beta_3 = 0.6979 \pm 0.0002$ [Percus and Martin, 1996]. For $d \geq 4$ no comparably pinned estimate exists, so we substitute the large-d asymptotic $\sqrt{d/(2\pi e)}$ [Bertsimas and van Ryzin, 1990], a form the same source proves for the minimum-spanning-tree and minimum-matching constants and only conjectures for the TSP constant. That substitution, together with $n \leq 1000$ at $d \geq 4$ falling short of the asymptotic regime, limits how far the $d \geq 4$ rows test the theorem itself.

F.2 Çavdar–Sokol

Çavdar–Sokol takes A as the area of the rectangle covering the nodes; $\bar{c}_j$ and cstdev_j are the mean and standard deviation of the absolute distances from the nodes to that rectangle's midpoint line on axis j, not to the sample mean. The divisor $\bar{c}_x \bar{c}_y$ sits inside the radical; placing it outside leaves the second term dimensionless. The estimator assumes an axis-aligned rectangle and is not rotation invariant, so we rotate each instance into its minimum-area enclosing rectangle by scanning convex-hull edges [Freeman and Shapira, 1975]. We then divide the raw estimate by the finite-n correction $0.9325e^{0.00005298n} - 0.2972e^{-0.01452n}$, Eq. (21) of Çavdar [2014], fitted over $n \in \{100, 125, \ldots, 975\}$ at $R^2 = 0.987$. Choi [2021] records that the uncorrected estimator underestimates for $n < 1000$, the direction this correction moves it. The correction is bounded to the range it was fitted over, and the bound binds on this benchmark: 1,320 of the 2,580 2D instances have $n < 100$, so for slightly over half the benchmark we evaluate the ratio at the $n = 100$ endpoint instead of extrapolating below it. The dissertation's own figures show the ratio continuing to fall toward 0.4 as n approaches 10, but publish no fitted form there, so extrapolating would substitute our curve for theirs. We hold the upper end fixed for the same reason: above $n = 975$ we apply no correction, because Eq. (21) grows without bound ($E/T = 1.22$ at $n = 5000$) and would contradict the training fit it exists to repair, leaving a 1.84% step at the upper boundary where the fitted ratio reads 0.982. The source regression targets Lin–Kernighan tour lengths, not optima, and every training graph carries a node at each corner of its rectangle, pinning the graph area to the covering rectangle in a way our instances do not.

F.3 Constant-Ratio References

The $\alpha = 1$ row returns L_{MST}. The asymptotic row uses $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}}$ for uniform planar points. The numerator is well established at 0.7124; the denominator is not. We quote 0.6331 following Cortina-Borja and Robinson [2000], a simulation estimate, not a theorem, and published estimates of the planar Euclidean MST constant span a range wider than its printed precision, so the ratio 1.1253 is approximate. The calibrated rows take the mean of α within each dimension, and within each dimension–size cell, over the training split only; the fitted values are frozen and released, not refitted at inference. At $d = 100$, which no training row covers, the table extrapolates from $d = 50$.

F.4 Hilbert Sort

The custom Hilbert sort independently min–max normalizes every axis to a 16-bit grid, sorts by a d-dimensional Hilbert index [Hilbert, 1891], visits nodes in that order, and closes the tour. This is a custom generalization inspired by the planar space-filling-curve heuristic of Bartholdi and Platzman [1982]; its planar performance guarantee does not transfer to our anisotropically normalized multidimensional ordering. It constructs a tour instead of estimating one, so its error is bounded below by the gap between a space-filling-curve tour and the optimum.

F.5 GART 1.0

GART 1.0's extractor builds a dense pairwise-distance matrix, so time and memory are $\Theta(n^2)$.

F.6 GART 2.0

GART 2.0's hybrid builder (Section 6) computes every feature the released model consumes. It stores bounding hypervolume as $\exp(\min(\log \mathrm{hv},\ 690))$, so the raw feature stays finite at high embedding dimension; the log-scale features are unaffected by the cap.

F.7 Convex-Hull Cap

The convex-hull plug-in uses the hull only for $d \leq 3$; above that we substitute the coordinate bounding box and record which was used. QHull's output size grows as $n^{\lfloor d/2 \rfloor}$, the same blow-up cited in Section 1 as a reason not to build estimators on hull geometry; at $d = 8$ with $n = 1000$ a single hull dominates the entire benchmark's runtime.

G Held–Karp Bound Ladders

G.1 Results on TSPLIB95 EUC_2D

We time GART 2.0 and the bound on the same instances as Table 17, each estimator alone in its own process during a quiet window. We compare the two on 77 of the 78 EUC_2D instances the bound's dense kernel covers; the one excluded instance is reported at the end of this subsection. The ms column is the median per-instance time, and the accuracy beside it is a MAPE and therefore a mean. $\times$ compares the bound's time with GART 2.0's: the ratio of the printed medians here and on the 2D and non-Euclidean ladders, and the median of back-to-back per-instance ratios on the multidimensional ladder. Raw is the incumbent $\hat{w}_k$. Cal. is $c_k \hat{w}_k$ with c_k fitted per budget on the planar training split. Bold marks every cell both cheaper and more accurate than GART 2.0. A *rung* is one budget row.

The calibrated bound dominates GART 2.0 on both cost and accuracy at an ascent budget of 25: 0.902 times the cost at 2.00% MAPE against 2.56%, on the same instances and protocol. The raw bound alone reaches parity at budget 50, at 1.46 times the cost and a margin of 0.00408 percentage points, with a paired win rate of exactly 50.0%. Both readings use the median of per-instance costs, the statistic the ladders report throughout; summed over the corpus instead, the budget-25 pair costs 24.9 times as much, because the bound is cheap on the many small instances that set the median and expensive on the few large ones that set the total. GART 2.0 therefore leaves the cost/accuracy front once the bound family is admitted, under the median convention.

On the 23 smallest instances the calibrated bound at that same budget costs 0.159 times GART 2.0 at 1.69% MAPE against 2.01%, and even the raw uncalibrated certificate reaches 22.2% lower error at 0.463 times the cost. In the middle bucket nothing dominates, and the raw bound needs the top of the ladder to match at all. In the largest bucket the raw bound matches at 6.65 times the cost and again nothing dominates. The reversal is monotone in n.

The bound is far more load-sensitive than GART 2.0: its median rises by a factor of 1.25 between a quiet and a noisy window at the crossing budget of 50, against 1.05 for GART 2.0. Every published repeat is therefore from one quiet window, with the noisy repeats retained only as a control; mixing the two would have overstated the bound's cost and flattered GART 2.0. On the excluded instance, `d18512`, above the bound's dense kernel switch, the bound costs 167 times GART 2.0 at the crossing budget and 2,078 times at the top of the ladder, 493 seconds against 237 milliseconds (the 239 ms of Section 4.8 is the same work under the harness protocol).

Table 14: TSPLIB95 EUC_2D cost/accuracy ladder over the 77 matched instances.

	$n \in [51, 150]$, $N = 23$				$n \in [151, 400]$, $N = 16$				$n > 400$, $N = 38$			
Row	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.
GART 2.0	3.73	1.00	2.01		4.55	1.00	2.38		18.3	1.00	2.97	
$k = 0$	0.195	0.0522	11.8	3.12	0.446	0.0979	11.3	3.59	24.5	1.34	9.13	4.01
$k = 10$	0.371	0.0996	8.47	2.04	0.911	0.200	9.85	3.76	43.9	2.40	7.63	2.53
$k = 25$	0.592	0.159	2.91	**1.69**	1.59	0.349	4.84	3.09	72.5	3.97	3.48	1.72
$k = 50$	0.958	0.257	2.09	**1.68**	2.79	0.614	3.82	3.03	122	6.65	2.31	1.56
$k = 100$	1.72	0.463	**1.56**	**1.20**	5.16	1.13	2.97	2.29	222	12.2	1.96	1.33
$k = 200$	3.22	0.863	**1.57**	**1.22**	10.1	2.22	2.96	2.28	428	23.4	1.88	1.26
$k = 500$	7.88	2.11	0.971	0.742	25.7	5.64	1.93	1.42	1,120	61.3	1.62	1.12

Table 15: 2D diverse cost/accuracy ladder, all 2,580 instances, both step rules.

	Volgenant–Jonker step				Polyak step			
Row	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.
GART 2.0	5.00 ms at $1.00\times$; 2.89% MAPE							
$k = 0$	0.230	0.0460	13.9	5.65	0.308	0.0616	13.9	5.65
$k = 10$	0.436	0.0872	8.35	4.26	0.492	0.0984	10.6	6.87
$k = 25$	0.704	0.141	3.61	**2.56**	0.729	0.146	7.52	6.39
$k = 50$	1.15	0.230	**2.70**	**2.33**	1.12	0.224	5.53	4.98
$k = 100$	2.05	0.409	**2.05**	**1.76**	1.88	0.376	3.19	**2.79**
$k = 200$	3.95	0.790	**1.88**	**1.62**	3.36	0.671	**1.88**	**1.65**
$k = 500$	9.42	1.88	1.27	1.10	7.15	1.43	1.45	1.33

G.2 Results on the 2D Diverse and Non-Euclidean Benchmarks

We timed the remaining two ladders the same way as the TSPLIB ladder of Section G.1, one estimator alone in its own process.

The raw bound passes GART 2.0 on the 2D diverse ladder (Table 15) under the Volgenant–Jonker step at a budget of 50, at 0.230 times the cost and a paired win rate of 54.8%; the calibrated row passes a rung earlier. The reversal is monotone in n and steeper than on TSPLIB (Section G.1), because this corpus reaches down to five nodes. On the 360 instances with $n \leq 10$, the raw bound at a budget of 25 reads 0.439% MAPE against GART 2.0's 3.21% at 0.0699 times the cost, winning 95.0% of the paired comparisons. On the 610 instances with $n > 500$, nothing dominates: the raw bound is more accurate only at the top budget, and there it costs 26.1 times as much.

On the non-Euclidean ladder (Table 16), the estimator needs the MDS embedding of Section 6 while the bound reads the distance matrix directly. Over the 29 instances both methods score, GART 2.0 costs 4.88 ms, of which the embedding is 9.21%, at 3.81% MAPE. The Polyak step at a budget of 500 reaches 0.509% MAPE at 0.693 times that cost, winning all 29 paired comparisons; at a budget of 200 it reads 0.762% for 0.339 times the cost, again on every instance. The cheapest domination by a bound row anywhere in this study is here: the Volgenant–Jonker step at a budget of 25 costs 0.0648 times GART 2.0 and reads 2.95% against 3.81%.

The two step rules do not order the same way on the two corpora. Volgenant–Jonker leads Polyak at every 2D budget that runs an ascent except 200, where the two are separated by half a hundredth of a percentage point; the gap is widest at a budget of 25, 3.61% against 7.52%. Polyak leads from a budget of 200 on the non-Euclidean corpus, 0.762% against 1.23%.

The timing sessions behind both ladders did not run on a quiet machine, and the two methods are not equally sensitive to that: a paired control against the published quiet-window measurements puts GART 2.0 at 1.48 times its published cost and the bound at 1.10 to 1.14 times, so every cost multiple above is between 0.740 and 0.768 of its quiet-box value. That direction runs against the bound: correcting for it costs exactly one domination, the raw Volgenant–Jonker row at a budget of 200 on the 2D corpus, whose 0.790 becomes 1.03. Running the released single-budget call once per rung instead of the checkpointed protocol, sharing nothing between rungs, costs 0.768 to 0.955 times the checkpointed ladder on the 2D corpus and 0.898 to 1.14 times on the non-Euclidean one.

H TSPLIB95 EUC_2D Benchmark by Size

Table 16: Non-EUC_2D TSPLIB95 ladder over the 29 instances both methods score, nine of which violate the triangle inequality; the planar c_k makes the Cal. columns indicative only.

	Volgenant–Jonker step				Polyak step			
Row	ms	×	Raw	Cal.	ms	×	Raw	Cal.
GART 2.0	4.88 ms at $1.00\times$; 3.81% MAPE							
$k = 0$	0.0327	0.00670	15.6	6.18	0.0518	0.0106	15.6	6.18
$k = 10$	0.170	0.0348	8.15	**3.19**	0.159	0.0327	10.1	5.48
$k = 25$	0.316	0.0648	**2.95**	**1.78**	0.280	0.0574	6.50	5.05
$k = 50$	0.556	0.114	**1.78**	**1.44**	0.488	0.100	3.82	**3.13**
$k = 100$	1.10	0.225	**1.27**	**1.04**	0.915	0.187	**1.99**	**1.59**
$k = 200$	2.15	0.441	**1.23**	**1.02**	1.66	0.339	**0.762**	**0.581**
$k = 500$	5.27	1.08	1.14	0.995	3.38	0.693	**0.509**	**0.492**

Table 17: TSPLIB95 EUC_2D benchmark; within each bucket rows are ordered by SDPE.

n range	Estimator	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)	R^2_α	Time (ms)
$n \in [51, 150]$ Count=23	**GART 2.0**	**2.63** [1.34, 3.58]	2.01	1.22	1.47	0.580	3.64
	L_{MST} ($\alpha = 1$)	3.77 [2.51, 4.50]	13.4	12.7	−13.4	−9.54	1.68
	Asymptotic MST ratio	4.24 [2.82, 5.07]	3.55	2.13	−2.58	−0.395	1.16
	Calibrated MST ratio $\hat{\rho}(d, n)$	4.68 [3.07, 5.68]	4.09	3.93	1.04	−0.176	1.14
	GART 1.0	5.64 [3.58, 7.19]	5.44	3.92	4.07	−1.31	4.05
	Custom Hilbert sort	11.5 [8.20, 13.9]	39.9	38.9	39.9	—	0.656
$n \in [151, 400]$ Count=16	**GART 2.0**	**2.86** [1.95, 3.51]	2.38	2.50	1.51	0.706	4.51
	L_{MST} ($\alpha = 1$)	4.82 [2.13, 6.65]	12.4	11.6	−12.4	−4.80	2.33
	Calibrated MST ratio $\hat{\rho}(d, n)$	5.37 [2.49, 7.39]	3.73	2.43	0.0110	0.0491	1.91
	Asymptotic MST ratio	5.42 [2.39, 7.48]	3.39	1.27	−1.41	−0.0879	1.84
	GART 1.0	5.53 [3.93, 6.54]	5.92	4.46	4.93	−0.497	5.03
	Custom Hilbert sort	7.35 [3.56, 10.4]	44.2	42.4	44.2	—	1.44
$n > 400$ Count=39	**GART 2.0**	**3.05** [2.32, 3.62]	2.90	1.98	2.68	−0.0710	20.5
	L_{MST} ($\alpha = 1$)	3.58 [2.84, 4.17]	9.74	9.39	−9.74	−6.45	9.55
	Calibrated MST ratio $\hat{\rho}(d, n)$	3.95 [3.13, 4.60]	3.54	3.24	1.75	−0.122	9.52
	Asymptotic MST ratio	4.03 [3.20, 4.70]	3.57	3.15	1.57	−0.132	9.27
	GART 1.0	7.64 [6.07, 8.92]	11.2	9.66	11.2	−10.7	18.4
	Custom Hilbert sort	12.6 [9.60, 15.7]	48.7	47.0	48.7	—	9.14
Total Count=78	**GART 2.0**	**2.92** [2.40, 3.38]	2.53	1.87	2.08	0.484	6.12
	L_{MST} ($\alpha = 1$)	4.21 [3.30, 4.99]	11.4	11.6	−11.4	−5.59	3.08
	Calibrated MST ratio $\hat{\rho}(d, n)$	4.48 [3.59, 5.32]	3.74	3.18	1.18	0.0872	2.62
	Asymptotic MST ratio	4.74 [3.72, 5.62]	3.53	2.67	−0.268	−0.0103	2.57
	GART 1.0	7.45 [6.24, 8.46]	8.43	6.51	7.80	−3.58	6.12
	Custom Hilbert sort	11.9 [9.86, 13.6]	45.2	44.6	45.2	—	2.64

I Trained Hyperparameters and Booster Statistics

A hyperparameter search does not improve on the frozen block enough to justify replacing it. Optuna's tree-structured Parzen estimator (TPE) ran 200 trials against in-distribution validation, 60 completed and 140 pruned; the tuned booster reaches 0.611% on the multidimensional test split against 0.623% for the frozen hyperparameters, and 2.64% against 2.47% on TSPLIB EUC_2D. Both boosters are unconstrained and trained on the logit target, so the monotone constraint of the released model does not enter the difference; scoring the tuned booster against the released model instead would confound the search with that constraint.

Table 18: GART 2.0 hyperparameters and trained-model statistics; the hyperparameter block is inherited and frozen.

Group	**Parameter**	**Value**
Hyperparameters (inherited, frozen, not tuned)		
Tree structure	num leaves	148
	min child samples	23
	feature fraction	0.548
	bagging fraction / freq	0.609 / 6
Optim. & reg.	learning rate	0.0260
	L1 regularization	7.76×10^{-2}
	L2 regularization	1.19×10^{-5}
Training configuration		
Training setup	boosting type	gbdt
	objective	regression_l2 (squared error)
	early-stopping patience	100
	random seed	42
Trained-booster statistics		
Booster	trees (after early stop)	1,118
	avg leaves / tree	147.80
	max leaves / tree	148
	avg tree depth	11.83
	max tree depth	40

Two other learners were fitted on the same feature pipeline before the gradient-boosted ensemble was frozen, and neither displaced it. A residual feed-forward network predicts the same α from the same MST-and-geometry descriptors, and on the 2D diverse benchmark it costs 138 to 146 ms per instance against GART 2.0's 103 to 148 ms, slower on five of the six generator classes and slower in aggregate. The network reads 2.98% MAPE overall against 2.89% and loses on five of the six classes, but it wins the near-collinear Line Noise class at 9.14% against 10.8% and finishes marginally ahead on aggregate dispersion, 4.61% SDPE against 4.68%. An ordinary least-squares fit is the cheapest of the three at 61.9 to 92.1 ms and pays for it in the tail, reaching 17.9% MAPE on the clustered class against 2.24%, eight times the released model's error, at $R^2_\alpha = -9.53$. Neither is a control on the released feature vector and we do not report them as one: the network was fitted on 30 columns and the linear model on 28 against the released block's 31, and neither was fitted on the released feature table.

J MDS Embedding for Non-Euclidean Instances

For TSPLIB ATT, GEO (geographic coordinates with great-circle distances), and EXPLICIT (distance matrix only) instances, classical MDS gives an approximate positive-eigenvalue Euclidean representation. CEIL_2D uses native coordinates. The embedding dimension κ is chosen by the rule of Section 6; this retained-mass rule is a dimension-selection heuristic, not a distance-preservation guarantee.

For GEO instances, the selected dimension ranges from 2 to 61; for screened EXPLICIT instances, from 10 to 100. The MST features (edge weights, degrees, diameter) are computed from the original distance matrix, while geometric features (centroid statistics, bounding volume, aspect ratio) use the approximate MDS coordinates.

Normalized distance stress is $\sqrt{\sum_{i<j}(D_{ij} - \hat{D}_{ij})^2 / \sum_{i<j} D_{ij}^2}$, where $\hat{D}_{ij}$ is the Euclidean distance induced by the retained MDS coordinates, and correlation is the Pearson correlation between the same distance vectors. Table 19 reports the median and range of each. CEIL_2D stress measures only integer-ceiling distortion, on a fixed-seed sample of 100,000–1,000,000 pairs; other rows use all pairs. Five capped instances do not reach the 99.9% target: `att532`, `pa561`, `si175`, `si535`, and `si1032`.

Table 19: Embedding diagnostics on the 23 screened instances; medians with [minimum, maximum]. "Below target": eigenvalue mass under 99.9% from the $\kappa = 100$ cap.

Distance type	N	κ range	Normalized stress	Distance correlation	Below target
CEIL_2D	4	2	0.0000 [0.0000, 0.0000]	1.00000 [1.00000, 1.00000]	0
ATT	2	6–100	0.0028 [0.0007, 0.0048]	0.99999 [0.99998, 1.00000]	1
GEO	10	2–61	0.0207 [0.0005, 0.2363]	0.99951 [0.96944, 1.00000]	0
EXPLICIT screened	7	10–100	0.0198 [0.0041, 0.1208]	0.99954 [0.99598, 0.99999]	4

The screened application set excludes ten EXPLICIT matrices with structural triangle-inequality violations, leaving seven, three of them with mild rounding-level violations. GART 2.0 declines one of them, `si1032`, and obtains 3.01% MAPE on the six it accepts. Distortion varies substantially within type, maximum stress 0.236 for GEO against 0.121 for screened EXPLICIT, so we do not attribute between-type accuracy differences to the distance label alone.